Justice-informed Cycling Infrastructure Upgrade via Multi-modal Bi-level Optimization

Manuel Campero Jurado[1], Carlos Canudas-de-Wit[2], Giovanni De Nunzio[3], Mauro Salazar[4]

***Abstract*—Promoting cycling requires an infrastructure that better protects cyclists from motor traffic, but reallocating road space to bicycles can increase car travel times. This creates a trade-off between reducing cyclists' exposure to motorized traffic and maintaining network efficiency, as well as a distributional question: Should interventions maximize aggregate benefits or prioritize users facing insufficient conditions?**

We address these questions through an integrated bi-level transport-network optimization framework for jointly designing cycling and car networks. It combines a safety-adjusted bicycle assignment model, a behavioral modal-shift model, and a machine-learning surrogate for car user equilibrium. We consider two justice-oriented formulations: a utilitarian model minimizing cyclist exposure and total system travel time, and a sufficientarian model reducing cycling-path insufficiency and trips exceeding a travel-time threshold. The former emphasizes aggregate performance, while the latter prioritizes users facing insufficient path-protection or travel-time conditions. Applied to Grenoble, both formulations reduce cyclist exposure while limiting increases in car travel times. The sufficientarian approach provides the most balanced outcome, demonstrating that path and travel-time sufficiency can be improved without sacrificing aggregate transport efficiency.

***Index Terms*—Multi-modal network optimization, transport justice, mode-shift modeling, bi-level optimization and sustainable transportation.**

I. Introduction

Cycling offers major health, environmental, and economic benefits, including improved public health, reduced congestion, and lower emissions [1], [2]. Consequently, several cities worldwide promote cycling as a sustainable transport mode, especially for short trips. However, safety and comfort remain key barriers to widespread adoption [3]. Physically separated cycling facilities have been shown to enhance safety and increase ridership [4], [5], yet reallocating road space to bicycles can reduce vehicle capacity and increase car travel times. This trade-off underscores the need for planning tools that improve cycling safety while balancing multi-modal traffic performance (see Fig. 1 for the existing bicycle and road networks in Grenoble).

From a societal perspective, cycling investments also raise distributional concerns. Building on transport justice theory [6], two perspectives are central: a utilitarian-efficiency approach that maximizes aggregate welfare by minimizing cyclist exposure and total system travel time, and a sufficientarian approach that seeks to reduce cycling-path insufficiency and trip insufficiency, i.e., cases where users fall below an acceptable exposure-based protection level or exceed a reasonable travel-time threshold. Incorporating these principles into network design requires models capable of capturing modal shifts, traffic equilibrium, and cycling-flow redistribution under infrastructure upgrades.

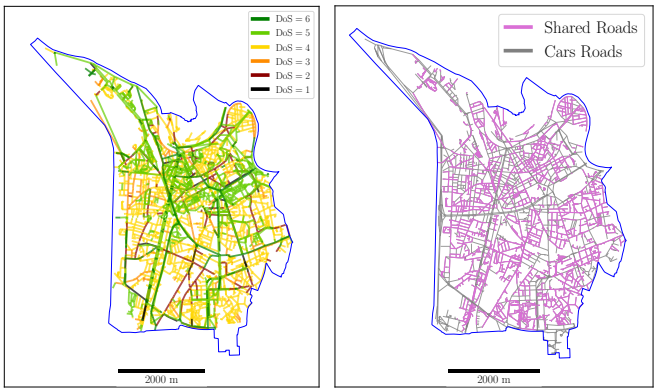


Fig. 1: Multi-modal transportation network of Grenoble, France. *Left:* Cycling infrastructure classified by degree of separation (DoS) of bike lanes, ranging from 1 (least separated) to 6 (fully separated). *Right:* Road network for cars, including links shared with bicycles.

This paper presents a framework for multi-modal network planning that integrates these perspectives under budget constraints. The model accounts for induced modal shifts from cars to bicycles following cycling infrastructure upgrades, dynamically adjusting travel demand across modes. To capture car traffic responses efficiently, we employ a surrogate machine learning model that approximates user equilibrium flows, allowing realistic yet computationally tractable city-scale analyses. Two bi-level optimization formulations are then introduced: a utilitarian model that minimizes aggregate cyclist exposure and total system travel time, and a sufficientarian model that maximizes the number of cyclists exceeding a path-sufficiency threshold while minimizing the proportion of cyclists and drivers who do not meet a maximum travel time threshold.

A. Literature Review

Our work lies at the intersection of three main research areas: cycling network optimization, modal-shift modeling, and transport justice.

Cycling network optimization: Research on cycling network planning has considered objectives such as infrastructure use,

[1]IFP Energies nouvelle, Rueil Malmaison, France, *manuel.campero-jurado@ifpen.fr* (*Corresponding author*)

[2]Univ. Grenoble Alpes, Inria, CNRS, Grenoble INP, GIPSA-Lab, 38000 Grenoble, France, *carlos.canudas-de-wit@gipsa-lab.fr*

[3]IFP Energies nouvelles, Rond-point de l'échangeur de Solaize, BP 3, 69360 Solaize, France, *giovanni.de-nunzio@ifpen.fr*

[4]Eindhoven University of Technology, 5600 MB, Eindhoven, The Netherlands, *m.r.u.salazar@tue.nl*

connectivity, safety-related conditions, investment efficiency, and interactions with other transport modes. Early studies formulated network design problems to increase the use of designated cycling facilities while improving connectivity and expand the cycling bike lanes [7], [8], [9]. Subsequent studies by Paulsen et al., Rich et al., and Consolini et al. examined benefit–cost trade-offs and developed budget-constrained methods to support cycling infrastructure investment decisions [10], [11], [12], [13]. More recent graph-based approaches have addressed cyclist protection by identifying or upgrading links toward higher levels of separation from motor traffic [14]. Multi-modal formulations have also considered the effects of cycling improvements on car travel times and traffic conditions [7], [8], [15], including the disruption caused by reallocating road capacity from cars to bicycles [16]. However, most approaches continue to emphasize aggregate measures of efficiency, connectivity, or exposure reduction, with limited attention to how the resulting benefits and burdens are distributed across users.

Modal shift in urban mobility: Cycling is increasingly recognized as a viable alternative to private cars for urban travel, although motorized transport continues to dominate road-space allocation and transport policies [17]. The quality and continuity of cycling infrastructure strongly influence its attractiveness. Cyclists generally prefer facilities that provide greater separation from motor traffic, whereas poor pavement quality and discontinuous routes can discourage cycling [18]. Consequently, connected and inclusive cycling networks are important for serving different users and trip purposes [19].

Behavioral and computational studies have examined how changes in infrastructure and travel conditions influence transport-mode choices. Agent-based models suggest that improved cycling conditions or reduced convenience for car travel can encourage shifts toward active mobility [20]. Empirical studies similarly indicate that many car trips could be replaced by active or low-carbon modes without substantial increases in travel time [21], [22]. Data-driven approaches have also been used to estimate modal-shift potential and the associated reduction in emissions [23], [24]. Nevertheless, modal-shift models are rarely integrated directly into cycling network optimization. In particular, existing approaches seldom examine how upgrades between different levels of separation from motor traffic jointly affect bicycle route choice and bicycle and car demand.

Transport justice: Traditional transport planning has largely followed a utilitarian perspective [25], seeking to maximize aggregate welfare or system efficiency without necessarily considering how benefits and burdens are distributed. Under this perspective, inequalities may be accepted when they lead to higher total utility [26], meaning that an intervention can be considered optimal even when some users receive limited benefits or experience worse conditions [27]. Alternative theories, including Rawlsian egalitarianism [28] and the capabilities approach, emphasize fair opportunities and the conditions experienced by disadvantaged users. Building on these ideas, Martens [6] proposes a sufficientarian perspective in which transport policy should guarantee a minimum acceptable level of accessibility. This principle has motivated the development of accessibility-sufficiency indicators [29], multi-modal applications [30], and capabilities-based analyses in different transport contexts [31]. Recent research has also distinguished between accessibility sufficiency, defined in terms of the destinations or opportunities that can be reached, and trip sufficiency, defined in terms of whether individual trips remain below an acceptable travel-time threshold [32]. Despite these developments, much of the transport-justice literature remains evaluative: it identifies inequalities or insufficient conditions but does not directly determine how networks should be redesigned. Recent studies have begun to incorporate justice principles into transport optimization [30], [32], [33], [34], but their integration with multi-modal infrastructure planning remains limited.

Overall, cycling network optimization, modal-shift modeling, and transport justice remain only partially connected. There is still a need for a prescriptive framework that jointly considers cyclist exposure, car-network performance, behavioral responses, and the distribution of path and trip sufficiency outcomes. Such a framework is needed to examine whether cycling investments can improve exposure-related conditions while maintaining travel-time efficiency and protecting users from inadequate trip conditions. In contrast to accessibility sufficiency, which is defined in terms of the number of destinations reachable within a reasonable time, this paper focuses on exposure-based path sufficiency and trip sufficiency. Accordingly, we use the term trip insufficiency for travel-time exceedance and reserve accessibility insufficiency for destination-based measures.

Statement of Contributions: This paper addresses this gap by proposing a multi-modal bi-level optimization framework for cycling infrastructure planning. Its overall contribution is to integrate cyclist exposure, car travel-time performance, modal shift, and distributive justice within a common network design framework. Two complementary formulations are considered: a utilitarian formulation that prioritizes aggregate exposure and travel-time efficiency, and a sufficientarian formulation that seeks to reduce exposure-based cycling-path insufficiency and trip insufficiency for bicycle and car users.

Preliminary versions of this work were presented at the 2025 IEEE Conference on Decision and Control [33] and the 2025 IEEE Intelligent Transportation Systems Conference [35]. The CDC paper compared utilitarian and sufficientarian formulations for cycling network optimization, focusing only on bicycle users. The ITSC paper showed that cycling infrastructure upgrades may increase car travel times when road capacity is reallocated. However, to the best of the authors' knowledge, no existing framework jointly accounts for cyclist exposure, car-network impacts, driver trip sufficiency, and modal shifts induced by cycling infrastructure upgrades. This extended framework evaluates cyclist exposure, car travel-time impacts, cycling-path sufficiency, and bicycle and driver trip insufficiency. It also adds a modal-shift component to account for changes in bicycle and car demand after cycling infrastructure upgrades. Bicycle flows are represented through a safety-adjusted shortest-path assignment, while car-network responses are approximated through a surrogate model for user equilibrium. Finally, the resulting design problems are solved

DoS	Description	F (veh/day)
6	Dedicated bike lanes, fully separated from motor traffic.	14 400
5	Bike lanes shared with pedestrians, separate from cars.	16 000
4	Mixed-traffic roads (30 km/h) without designated bike lanes.	960
3	Shared roads with cycling pictograms.	3 120
2	Roads with painted bike lanes.	6 912
1	Unmarked shared car-bike lanes.	8 448

TABLE I: The degree of separation (DoS) categorizes bike lane infrastructure based on safety levels. The third column presents annual average daily traffic (denoted as F) on adjacent car roads, sourced from the GTLVille platform in Grenoble [36].

using a genetic algorithm and a modified connectivity-based heuristic and evaluated on the Grenoble, France, network. The case study illustrates how the integrated framework can compare utilitarian and sufficientarian planning strategies under multi-modal interactions and induced modal shifts.

II. Model Design

In this section, we outline the methodological framework of the study. We begin by introducing the classification of bike lanes and the construction of the multi-modal network graph. We then present the cyclist flow prediction model and the car flow model, followed by their interaction within the shared network. On this basis, we formulate two bi-level optimization problems. The first adopts a utilitarian perspective, aiming to minimize cyclists' overall exposure to motor traffic while mitigating the increase in car travel times. The second follows a sufficientarian perspective, seeking to reduce exposure-based path insufficiency and trip insufficiency for bicycle and car users, again while accounting for the impact on car travel times.

A. Bike Lane Safety Classification

The Degree of Separation (DoS) classification, adapted from an OSM-based platform, categorizes cycling infrastructure based on separation from motor traffic. Table I presents DoS categories, showing that higher-DoS lanes offer greater safety. These scores serve as weights in the network graph, modeling upgrades from lower to higher safety levels. The classification methodology is detailed in [14], [37]. Fig. 1 illustrates the spatial distribution of these categories alongside the motorized road network in the Grenoble metropolitan area.

B. Graph Representation

The transportation network is modeled as a weighted directed graph (digraph) $\mathcal{G} = (\mathcal{V}, \mathcal{A}, \mathcal{W})$, where $\mathcal{V}$ is the set of nodes (e.g., intersections or road junctions), $\mathcal{A}$ is the set of directed arcs (or links) representing road segments, and $\mathcal{W}$ denotes the corresponding set of arc weights representing a safety factor.

The network comprises two distinct but overlapping layers: the bicycle layer $\mathcal{G}^{\mathrm{b}} = (\mathcal{V}^{\mathrm{b}}, \mathcal{A}^{\mathrm{b}}, \mathcal{W}^{\mathrm{b}})$ and the car layer $\mathcal{G}^{\mathrm{c}} = (\mathcal{V}^{\mathrm{c}}, \mathcal{A}^{\mathrm{c}}, \mathcal{W}^{\mathrm{c}})$. Each layer has its own set of nodes, arcs, and weights. Specifically, for each bike arc $a = (u, v) \in \mathcal{A}^{\mathrm{b}}$, the weight w_a^{b} represents the DoS of a, and is symmetric: $w_{(u,v)}^{\mathrm{b}} = w_{(v,u)}^{\mathrm{b}}$. In contrast, car arcs $a \in \mathcal{A}^{\mathrm{c}}$ are uniformly weighted with $w_a^{\mathrm{c}} = t_{a,0}^{\mathrm{c}}$, where $t_{a,0}^{\mathrm{c}}$ denotes the travel time required to traverse arc a under uncongested conditions in the original, unmodified road network. Each arc also has an associated geographical length, denoted d_a^{b} for bicycles and d_a^{c} for cars, respectively. The full multi-modal network satisfies the union relations $\mathcal{V} = \mathcal{V}^{\mathrm{b}} \cup \mathcal{V}^{\mathrm{c}}$ and $\mathcal{A} = \mathcal{A}^{\mathrm{b}} \cup \mathcal{A}^{\mathrm{c}}$. Throughout the modeling we use the same notation a for arcs in $\mathcal{A}^{\mathrm{b}}$ and $\mathcal{A}^{\mathrm{c}}$ to simplify expressions, as some arcs belong to both layers ($\mathcal{A}^{\mathrm{b}} \cap \mathcal{A}^{\mathrm{c}} \neq \emptyset$). The layer is indicated by superscripts b or c when needed. Finally, let $\mathcal{V}_{\mathcal{O}}^{\mathrm{b}}, \mathcal{V}_{\mathcal{D}}^{\mathrm{b}} \subseteq \mathcal{V}^{\mathrm{b}}$ denote the sets of origin and destination nodes for bicycle trips, and similarly $\mathcal{V}_{\mathcal{O}}^{\mathrm{c}}, \mathcal{V}_{\mathcal{D}}^{\mathrm{c}} \subseteq \mathcal{V}^{\mathrm{c}}$ for car trips.

C. Cyclist Exposure

For $(u, v) \in \mathcal{V}_{\mathcal{O}}^{\mathrm{b}} \times \mathcal{V}_{\mathcal{D}}^{\mathrm{b}}$, let $q_{(u,v)}^{\mathrm{b}}$ denote the cycling demand between nodes u and v. Following [7], we adopt an all-or-nothing (AoN) assignment for bicycles, assuming negligible congestion effects as in [38]. Under this model, the entire demand $q_{(u,v)}^{\mathrm{b}}$ is assigned to the shortest path between u and v, where distance is measured using a cyclist-specific metric that accounts for the topological and environmental characteristics of the cycling infrastructure.

The spatial distribution of bicycle flows is obtained through a procedure analogous to betweenness centrality [39], which determines whether an arc lies on the shortest path between any origin–destination pair and, under the AoN assumption, assigns the corresponding demand to that path. For an arc $a \in \mathcal{A}^{\mathrm{b}}$ belonging to the shortest path $p_{(u,v)}^{\mathrm{b}} = (a_1, \ldots, a_k)$, d_a^{b} denotes its geographical length; $\measuredangle_a^{\mathrm{b}}$ the angular change (in degrees) if it is the first or last arc in the path, or at junctions between consecutive arcs; s_a^{b} the slope factor based on the terrain slope m_a^{b}; and f_a^{b} the traffic factor, representing the influence of adjacent car traffic on perceived cycling comfort and safety.

The cyclist-specific distance $\bar{d}_{(u,v)}^{\mathrm{b}}$ between u and v, as defined in [38], is then expressed as

$$\bar{d}_{(u,v)}^{\mathrm{b}} = \sum_{a \in p_{(u,v)}^{\mathrm{b}}} \left(d_a^{\mathrm{b}} \cdot s_a^{\mathrm{b}} \cdot (f_a^{\mathrm{b}})^{\rho} \right) + \sum_{a \in p_{(u,v)}^{\mathrm{b}}} \measuredangle_a^{\mathrm{b}} \cdot \kappa, \qquad (1)$$

where $\rho > 0$ and $\kappa > 0$ are calibration parameters. The term f_a^{b} represents the traffic exposure factor on arc a and is defined as

$$f_a^{\mathrm{b}} = \begin{cases} \psi \cdot \exp\left(F(a)\right), & \text{if } w_a^{\mathrm{b}} < 5, \\ \psi \cdot 2.03, & \text{if } w_a^{\mathrm{b}} = 5, \\ \psi, & \text{if } w_a^{\mathrm{b}} = 6, \end{cases} \qquad (2)$$

where ψ is a scaling factor, w_a^{b} denotes the DoS of bike lane a, and $F(a)$ is the annual average daily traffic on adjacent car lanes, as reported in Table I. According to [40], the effect of AADT becomes negligible for the fully separated cycling infrastructure (DoS $= 5$ or 6). The factor 2.03 reflects the more than twofold increase in crash risk in pedestrian-shared spaces compared to fully separated facilities [41]. For convenience,

we also define the cyclist-specific distance contributed by a single arc $a \in p^{\mathrm{b}}_{(u,v)}$ as

$$\bar{d}^{\mathrm{b}}_a = d^{\mathrm{b}}_a \cdot s^{\mathrm{b}}_a \cdot (f^{\mathrm{b}}_a)^{\rho} + \measuredangle^{\mathrm{b}}_a \cdot \kappa. \tag{3}$$

This arc-level quantity is introduced only to express the path-level cyclist-specific distance in additive form, namely $\bar{d}^{\mathrm{b}}_{(u,v)} = \sum_{a \in p^{\mathrm{b}}_{(u,v)}} \bar{d}^{\mathrm{b}}_a$. It should therefore be interpreted as the contribution of arc a to the generalized cyclist distance used for route choice, rather than as an independently validated travel-time or safety measure. The calibration and validation of the underlying path-level metric follow [38].

Unlike traditional car traffic assignment, where link flows are obtained by solving a user equilibrium problem to reflect congestion and travel-time feedback, bicycle flows are determined directly by an all-or-nothing (AoN) assignment induced by the safety-adjusted cyclist distance $\bar{d}^{\mathrm{b}}_{(u,v)}$. This assumes that cyclists select the most attractive (shortest) route in terms of perceived distance and safety, without mutual interaction effects. Specifically, let $\mathbb{I}_{a \in p^{\mathrm{b}}_{(u,v)}}$ be an indicator equal to 1 if arc a belongs to the shortest path $p^{\mathrm{b}}_{(u,v)}$ between OD pair (u, v) and 0 otherwise. The induced AoN link flow is

$$\phi^{\mathrm{b}}_a = \sum_{(u,v) \in \mathcal{V}^{\mathrm{b}}_{\mathcal{O}} \times \mathcal{V}^{\mathrm{b}}_{\mathcal{D}}} \tilde{q}^{\mathrm{b}}_{(u,v)} \cdot \mathbb{I}_{a \in p^{\mathrm{b}}_{(u,v)}} \qquad \forall a \in \mathcal{A}^{\mathrm{b}}. \tag{4}$$

To obtain a network-wide exposure measure that remains comparable across scenarios (e.g., when total cycling demand changes due to mode shift), we define the overall cyclist exposure as the demand-normalized link-based exposure:

$$J_{\mathcal{E}} = \sum_{a \in \mathcal{A}^{\mathrm{b}}} \phi^{\mathrm{b}}_a \cdot f^{\mathrm{b}}_a \cdot d^{\mathrm{b}}_a. \tag{5}$$

D. Cars Travel Time

Similarly to the bicycle network, let $\tilde{q}^{\mathrm{c}}_{(u,v)}$ denote the demand between nodes $(u, v) \in \mathcal{V}^{\mathrm{c}}_{\mathcal{O}} \times \mathcal{V}^{\mathrm{c}}_{\mathcal{D}}$. To distinguish the exact car-assignment model from the surrogate approximation introduced later, we use quantities with a tilde to denote values obtained from the exact user-equilibrium assignment, such as $\tilde{\boldsymbol{\phi}}^{\mathrm{c}}$ and $\tilde{\mathcal{T}}_{\mathrm{c}}$, while the corresponding non-tilded quantities denote surrogate-based approximations used inside the upper-level optimization. In contrast to the all-or-nothing traffic assignment, we consider the user equilibrium traffic assignment principle, as described in [42], which states that no driver can unilaterally reduce their travel time by switching to another route. To compute this equilibrium, it is essential to account for capacity constraints: dedicating a lane to bicycles reduces the effective capacity for cars. This reduction depends on the number of existing car lanes and the physical feasibility of the reallocation. To distinguish the exact car-assignment model from the surrogate approximation introduced later, quantities with a tilde, such as $\tilde{\boldsymbol{\phi}}^{\mathrm{c}}$ and $\tilde{\mathcal{T}}_{\mathrm{c}}$, denote values obtained from the exact user-equilibrium assignment, whereas the corresponding quantities without a tilde denote surrogate-based approximations used inside the upper-level optimization.

Let ϕ^{c}_a denote the flow on car link a, and let $t^{\mathrm{c}}_a(\phi^{\mathrm{c}}_a)$ be the corresponding travel time, which is a function of ϕ^{c}_a. This travel time is modeled using the Bureau of Public Roads (BPR) function [43], given by

$$t^{\mathrm{c}}_a(\phi^{\mathrm{c}}_a) = t^0_a \left(1 + \mu \left(\frac{\phi^{\mathrm{c}}_a}{\theta^{\mathrm{c}}_a}\right)^{\sigma}\right), \tag{6}$$

where t^0_a is the free-flow travel time on link a, θ^{c}_a is the flow capacity of the link a, and $\mu = 0.15$, $\sigma = 4$ are calibrated parameters. Although several studies have shown that the BPR formulation may not always accurately capture congestion dynamics under all conditions, it remains widely adopted in network modeling due to its simplicity, interpretability, and compatibility with standard traffic assignment methods.

To solve the user equilibrium (UE) assignment, that is, to determine the flow ϕ^{c}_a for each link, for each pair of nodes $(u, v) \in \mathcal{V}^{\mathrm{c}}_{\mathcal{O}} \times \mathcal{V}^{\mathrm{c}}_{\mathcal{D}}$, let $\mathcal{P}_{(u,v)}$ denote the set of possible car paths from node u to node v, and let $\phi^{p^{\mathrm{c}}}_{(u,v)}$ be the flow assigned to path $p^{\mathrm{c}} \in \mathcal{P}_{(u,v)}$. We impose the following demand constraint:

$$\sum_{p^{\mathrm{c}} \in \mathcal{P}_{(u,v)}} \phi^{p^{\mathrm{c}}}_{(u,v)} = \tilde{q}^{\mathrm{c}}_{(u,v)}. \tag{7}$$

Next, the total flow on each arc $a \in \mathcal{A}^{\mathrm{c}}$ is computed by aggregating the path flows in the following way:

$$\phi^{\mathrm{c}}_a = \sum_{(u,v) \in \mathcal{V}^{\mathrm{c}}_{\mathcal{O}} \times \mathcal{V}^{\mathrm{c}}_{\mathcal{D}}} \sum_{p^{\mathrm{c}} \in \mathcal{P}_{(u,v)}} \phi^{p^{\mathrm{c}}}_{(u,v)} \cdot \mathbb{I}_{a \in p^{\mathrm{c}}}, \tag{8}$$

where $\mathbb{I}_{a \in p^{\mathrm{c}}}$ is an indicator function equal to 1 if arc a is part of path p^{c}, and 0 otherwise. Additionally, we require non-negativity of the path and arcs flows

$$\phi^{p^{\mathrm{c}}}_{(u,v)} \geq 0, \quad \forall p^{\mathrm{c}} \in \mathcal{P}_{(u,v)}, \quad \forall (u, v) \in \mathcal{V}^{\mathrm{c}}_{\mathcal{O}} \times \mathcal{V}^{\mathrm{c}}_{\mathcal{D}}, \tag{9}$$

$$\phi^{\mathrm{c}}_a \geq 0, \quad \forall a \in \mathcal{A}^{\mathrm{c}}. \tag{10}$$

The car total system travel time associated with a feasible car-flow vector is defined as

$$\tilde{\mathcal{T}}_{\mathrm{c}} = \sum_{a \in \mathcal{A}^{\mathrm{c}}} \phi^{\mathrm{c}}_a \cdot t^{\mathrm{c}}_a(\phi^{\mathrm{c}}_a). \tag{11}$$

This quantity measures the aggregate travel time experienced by drivers. To compute the user-equilibrium assignment, we instead minimize the Beckmann potential function [44]. Let $\boldsymbol{\phi}^{\mathrm{c}} = [\phi^{\mathrm{c}}_a]^{\top}_{a \in \mathcal{A}^{\mathrm{c}}} \in \mathbb{R}^{|\mathcal{A}^{\mathrm{c}}|}_{\geq 0}$ denote the vector of car-link flows. The user-equilibrium traffic assignment problem is then formulated as the following optimization program:

Problem 1 (User Equilibrium Traffic Assignment):

$$\boldsymbol{\phi}^{\mathrm{c}\star} = \arg \min_{\boldsymbol{\phi}^{\mathrm{c}} \in \mathbb{R}^{|\mathcal{A}^{\mathrm{c}}|}_{\geq 0}} \sum_{a \in \mathcal{A}^{\mathrm{c}}} \int_0^{\phi^{\mathrm{c}}_a} t^{\mathrm{c}}_a(x)\, \mathrm{d}x$$

$$\text{s.t.} \quad (7)\text{–}(10).$$

Then, $\phi^{c^\star} = \left[\phi^{c^\star}_a\right]^{\top}_{a \in \mathcal{A}^{\mathrm{c}}} \in \mathbb{R}^{|\mathcal{A}^{\mathrm{c}}|}_{\geq 0}$ is the vector of optimal car flows such that no driver can unilaterally reduce their travel time by choosing an alternative route; this corresponds to the user equilibrium condition. The convexity of $\mathcal{T}$ and the fact that every minimizer $\boldsymbol{\phi}^{c^\star}$ of Problem 1 corresponds to a user equilibrium, i.e., a Nash equilibrium, can be found in [45]. In the subsequent section, we proceed to integrate the bicycle traffic exposure model into the car traffic assignment model based on travel time. For cyclists, we recall that the network

assignment relies on the safety-adjusted cyclist distance $\bar{d}_a^{\mathrm{b}}$, which accounts for geometric, slope, and traffic-related factors on each arc. Let $\tilde{\boldsymbol{w}}^{\mathrm{b}} = \left[\tilde{w}_a^{\mathrm{b}}\right]_{a\in\mathcal{A}^{\mathrm{b}}}^{\top}$ denotes the vector representing the DoS of all bicycle lanes, and let $\boldsymbol{w}^{\mathrm{b}} = \left[w_a^{\mathrm{b}}\right]_{a\in\mathcal{A}^{\mathrm{b}}}^{\top}$ represent the corresponding upgraded configuration, where DoS values have been increased. By definition, each element satisfies $\forall a \in \mathcal{A}^{\mathrm{b}} : \tilde{w}_a^{\mathrm{b}} \le w_a^{\mathrm{b}} \le 6$. To assess accessibility and the temporal efficiency of the cycling network, distances can be converted into travel times using a representative average cycling speed v_b. Accordingly, the travel time along a bicycle path $p_{(u,v)}^{\mathrm{b}}$ between an origin–destination pair (u, v) is given by

$$t_{p_{(u,v)}}^{\mathrm{b}}(\boldsymbol{w}^{\mathrm{b}}) \;=\; \sum_{a\in p_{(u,v)}^{\mathrm{b}}} \frac{\bar{d}_a^{\mathrm{b}}(w_a^{\mathrm{b}})}{v_b}. \tag{12}$$

The bicycle total system travel time in the network is then obtained by summing the travel times of all origin–destination pairs.

$$\mathcal{T}_{\mathrm{b}}(\boldsymbol{w}^{\mathrm{b}}) \;=\; \sum_{(u,v)\in\mathcal{V}_{\mathcal{O}}^{\mathrm{b}}\times\mathcal{V}_{\mathcal{D}}^{\mathrm{b}}} \phi_{(u,v)}^{\mathrm{b}}(\boldsymbol{w}^{\mathrm{b}}) \cdot t_{p_{(u,v)}}^{\mathrm{b}}(\boldsymbol{w}^{\mathrm{b}}). \tag{13}$$

Finally, the total system travel time, combining both car and bicycle flows, is expressed as

$$\tilde{\mathcal{T}}_{\mathrm{tot}}(\boldsymbol{w}^{\mathrm{b}}) \;=\; \tilde{\mathcal{T}}_{\mathrm{c}}(\boldsymbol{w}^{\mathrm{b}}) \;+\; \mathcal{T}_{\mathrm{b}}(\boldsymbol{w}^{\mathrm{b}}). \tag{14}$$

Here, the tilde in $\tilde{\mathcal{T}}\mathrm{tot}$ indicates that the car component is computed from the exact user-equilibrium assignment. The corresponding surrogate-based total travel time, introduced later, is denoted without a tilde.

E. Interaction Between Cars and New Bike Lanes

Upgrading a bike lane may reduce the capacity of the adjacent car lane. However, as noted by [46], bike lanes that are not fully separated, specifically those with a degree of separation (DoS) less than or equal to 4, do not significantly affect car travel times. In our framework, only upgrades to a DoS of at least 5 (i.e., fully separated infrastructure) are assumed to impact car traffic flow. The effective capacity of a car link, denoted θ_a^{c}, depends on its original capacity $\tilde{\theta}_a^{\mathrm{c}}$, the number of car lanes n_a^{c}, the road width l_a, and the upgraded bike lane DoS w_a^{b}. To simplify the notation, we define a binary indicator $z_a \in \{0, 1\}$, where $z_a = 1$ if the bike lane on link a is upgraded to a fully separated facility (i.e., $\tilde{w}_a^{\mathrm{b}} \ge 5$) on a narrow road with l_a not exceeding a given width threshold, and $z_a = 0$ otherwise. The effective capacity is then written compactly as

$$\theta_a^{\mathrm{c}} = \tilde{\theta}_a^{\mathrm{c}}\left(1 - \frac{z_a}{n_a^{\mathrm{c}}}\right). \tag{15}$$

This formulation applies a capacity reduction only when a bike lane is upgraded to a fully separated infrastructure (DoS ≥ 5). The reduction is scaled by the number of existing car lanes through the factor $\frac{1}{n_a^{\mathrm{c}}}$, so that reallocating one lane has a larger relative effect on single-lane roads than on multi-lane roads. This provides a simple way to capture the asymmetric impact of bike-lane upgrades on car capacity. Since such upgrades may increase car travel times while also encouraging some users to shift from cars to bicycles, we next introduce the mode-choice model.

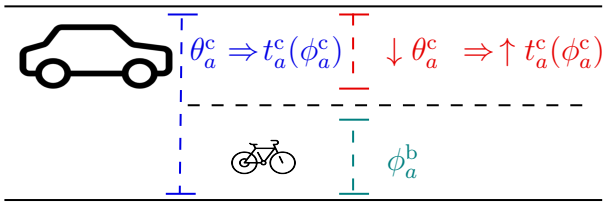


Fig. 2: The creation/upgrading of a bike lane is modeled as reducing the capacity of the affected car link, resulting in an increased travel time $\uparrow t_a^{\mathrm{c}}(\phi_a^{\mathrm{c}})$.

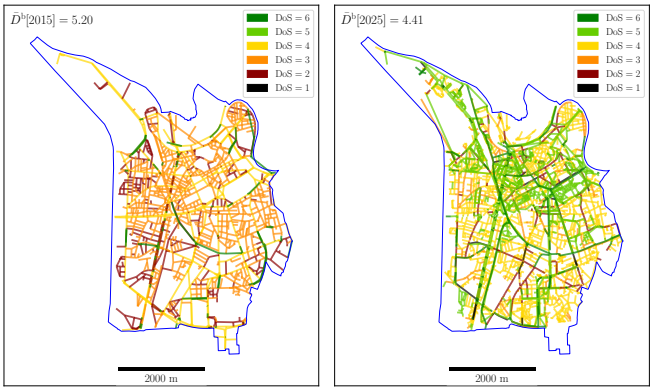


Fig. 3: Historical cycling network of Grenoble, France. *Left:* year 2015. *Right:* year 2025. The average cycling distance 2 $\overline{D}^{\mathrm{b}}[k]$ decreases as the network expands and improves.

We develop a binary mode choice model linking changes in cycling network quality to variations in bicycle mode share over time, using historical data from the Grenoble metropolitan area. The annual cycling networks for 2015–2025 were extracted from OpenStreetMap using the `OSMnx` Python package [47]. For each year $k \in \{2015, 2016, \ldots, 2025\}$, we compute the average (safety-adjusted) cycling distance as

$$\overline{D}^{\mathrm{b}}[k] = \frac{1}{|\mathcal{V}_{\mathcal{O}}^{\mathrm{b}} \times \mathcal{V}_{\mathcal{D}}^{\mathrm{b}}|} \sum_{(u,v)\in\mathcal{V}_{\mathcal{O}}^{\mathrm{b}}\times\mathcal{V}_{\mathcal{D}}^{\mathrm{b}}} \bar{d}_{(u,v)}^{\mathrm{b}}[k], \tag{16}$$

where $\bar{d}_{(u,v)}^{\mathrm{b}}[k]$ denotes the cyclist distance between origin u and destination v for the network configuration and traffic conditions of year k. Fig. 3 illustrates the Grenoble cycling network in 2015 and its projected configuration for 2025. Fig. 4 reports the average cycling distance and the bicycle mode share for the historical network between 2015 and 2025, based on observed values from French National Institute of Statistics and Economic Studies (INSEE) and other surveys [48], [49], [50], with intermediate years estimated through linear regression. As shown, there is an inverse relationship between the two variables: the average cycling distance decreases almost monotonically over time, while the bicycle mode share steadily increases, reflecting a progressive shift toward shorter but more frequent cycling trips as infrastructure expands and connectivity improves.

We acknowledge that cycling adoption is influenced by a wide range of factors, including social context, municipal initiatives, historical development, and gender dynamics [51],

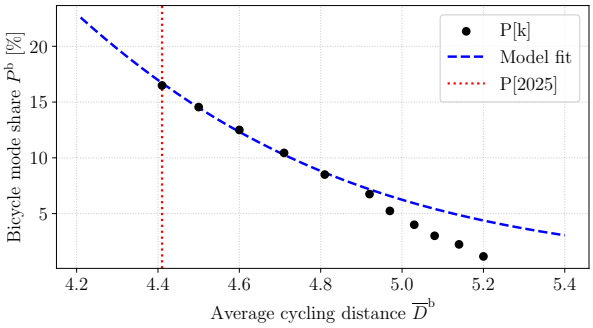


Fig. 4: The logit model plot to predict $P^{\mathrm{b}}[k]$ using $\bar{D}^{\mathrm{b}}$.

[52]. However, in this study we adopt a macroscopic formulation in which infrastructure is the sole explanatory variable of bicycle mode choice. This assumption is supported by evidence showing that the quality of cycling facilities strongly affects travel behavior, as cyclists consistently prefer routes separated from motorized traffic, while poor surface conditions tend to discourage cycling [18], [20]. We model the annual probability of choosing cycling, denoted as $P^{\mathrm{b}}[k]$, based on the utility associated with the bicycle mode, $V^{\mathrm{b}}[k]$. This utility is assumed to depend linearly on the average cycling distance $\overline{D}^{\mathrm{b}}[k]$

$$V^{\mathrm{b}}[k] = \alpha + \beta\, \overline{D}^{\mathrm{b}}[k]. \tag{17}$$

Using a binary logit model, with the car mode as the reference alternative, the probability of choosing the bicycle mode in year k is expressed as

$$P^{\mathrm{b}}[k] = \frac{\exp(V^{\mathrm{b}}[k])}{\exp(V^{\mathrm{b}}[k]) + \exp(V^{\mathrm{c}}[k])}. \tag{18}$$

Since only relative utilities influence the choice, we normalize the utility of the car mode to zero ($V^{\mathrm{c}} \equiv 0$) without loss of generality. This yields the standard logit formulation

$$P^{\mathrm{b}}[k] = \frac{\exp(V^{\mathrm{b}}[k])}{1 + \exp(V^{\mathrm{b}}[k])} = \frac{1}{1 + \exp(-V^{\mathrm{b}}[k])}, \tag{19}$$

where α is an alternative-specific constant capturing unobserved factors favoring car use (e.g., travel time, fuel cost, or parking convenience), and $\beta < 0$ represents the sensitivity of the cycling adoption to the perceived distance or effort. Substituting (17) into the logit expression (19) yields the predicted bicycle mode share

$$\widehat{P}^{\mathrm{b}}[k] = \frac{1}{1 + \exp\big(-(\alpha + \beta\, \overline{D}^{\mathrm{b}}[k])\big)}. \tag{20}$$

The parameters (α, β) are estimated from the observed annual data pairs $\{(\overline{D}^{\mathrm{b}}[k], P^{\mathrm{b}}[k])\}$ by minimizing the squared difference between observed and predicted bicycle shares:

$$\min_{\alpha,\beta} \sum_{k} \Big[P^{\mathrm{b}}[k] - \widehat{P}^{\mathrm{b}}[k]\Big]^2. \tag{21}$$

To ensure consistency with the current situation, α is calibrated so that, for the existing 2025 network, the model reproduces the observed cycling share. Hence, if no network upgrades are applied (*i.e.*, $\boldsymbol{w}^{\mathrm{b}}$ identical to 2025), the predicted bicycle probability $\mathbb{P}^{\mathrm{b}}$ equals the observed value P^{b}_{2025}.

For any upgraded configuration $\boldsymbol{w}^{\mathrm{b}}$, the corresponding average cycling distance $\overline{D}^{\mathrm{b}}(\tilde{\boldsymbol{w}}^{\mathrm{b}})$ determines the updated mode shares. Although $\overline{D}^{\mathrm{b}}[k]$ varies over time, it can also be viewed as a function of the network configuration $\boldsymbol{w}^{\mathrm{b}}$, since infrastructure upgrades that increase the DoS, modify travel effort and route choices, thus affecting the resulting cycling share.

$$\mathbb{P}^{\mathrm{b}}(\boldsymbol{w}^{\mathrm{b}}) = \frac{1}{1 + \exp\big(-(\alpha + \beta\, \overline{D}^{\mathrm{b}}(\boldsymbol{w}^{\mathrm{b}}))\big)}, \tag{22}$$

$$\mathbb{P}^{\mathrm{c}}(\boldsymbol{w}^{\mathrm{b}}) = 1 - \mathbb{P}^{\mathrm{b}}(\boldsymbol{w}^{\mathrm{b}}), \tag{23}$$

and the total travel demand between each origin–destination pair (u, v) is reallocated as

$$q^{\mathrm{b}}_{(u,v)}(\boldsymbol{w}^{\mathrm{b}}) = \mathbb{P}^{\mathrm{b}}(\boldsymbol{w}^{\mathrm{b}}) \cdot Q_{(u,v)}, \tag{24}$$

$$q^{\mathrm{c}}_{(u,v)}(\boldsymbol{w}^{\mathrm{b}}) = \mathbb{P}^{\mathrm{c}}(\boldsymbol{w}^{\mathrm{b}}) \cdot Q_{(u,v)}, \tag{25}$$

where $Q_{(u,v)}$ denotes the total travel demand of car drivers and cyclists between origin u and destination v.

1) Surrogate-Based Flow Prediction with Feasibility Projection: Computing the UE traffic assignment at each iteration is computationally expensive, particularly when travel demand and lane configurations change frequently due to network upgrades. To mitigate this cost, we employ a graph-based surrogate that predicts near-equilibrium car flows and travel times directly from network-level features, followed by a projection step to ensure feasibility.

Let $\tilde{\boldsymbol{\phi}}^{\mathrm{c}} = [\tilde{\phi}^{\mathrm{c}}_a]^\top_{a\in\mathcal{A}^{\mathrm{c}}} \in \mathbb{R}^{|\mathcal{A}^{\mathrm{c}}|}$ denote the vector of car flows on all arcs, and let $\tilde{\boldsymbol{q}}^{\mathrm{c}} = [\tilde{q}^{\mathrm{c}}_{(u,v)}]^\top_{(u,v)\in\mathcal{V}^{\mathrm{b}}_{\mathcal{O}}\times\mathcal{V}^{\mathrm{b}}_{\mathcal{D}}} \in \mathbb{R}^{|\mathcal{V}^{\mathrm{c}}_{\mathcal{O}}\times\mathcal{V}^{\mathrm{c}}_{\mathcal{D}}|}$ represent the OD demand vector. The UE flow $\boldsymbol{\phi}^{\mathrm{c}^\star}$ satisfies the flow-conservation constraint

$$B\tilde{\boldsymbol{\phi}}^{\mathrm{c}^*} = \boldsymbol{d}^{\mathrm{c}}, \tag{26}$$

where $B \in \{-1, 0, 1\}^{|\mathcal{V}^{\mathrm{c}}|\times|\mathcal{A}^{\mathrm{c}}|}$ is the node–arc incidence matrix and $\boldsymbol{d}^{\mathrm{c}} \in \mathbb{R}^{|\mathcal{V}^{\mathrm{c}}|}$ is the net demand vector (positive at origins, negative at destinations, and zero at intersections and junctions).

We train a Graph Convolutional Network (GCN) surrogate $\mathcal{M}$ on UE traffic assignments obtained under random bike-lane upgrade configurations $\tilde{\boldsymbol{w}}^{\mathrm{b}}$. Each configuration alters the car network capacities and OD demands, generating distinct equilibrium flows and travel times used as ground truth. The GCN takes as input the car network topology $\mathcal{A}^{\mathrm{c}}$, the effective arc capacities $\boldsymbol{\theta}^{\mathrm{c}} = [\theta^{\mathrm{c}}_a]_{a\in\mathcal{A}^{\mathrm{c}}}$ from (15), and the post-shift OD demand vector $\boldsymbol{q}^{\mathrm{c}} = [q^{\mathrm{c}}_{(u,v)}]^\top_{(u,v)\in\mathcal{V}^{\mathrm{c}}_{\mathcal{O}}\times\mathcal{V}^{\mathrm{c}}_{\mathcal{D}}}$ from (24), and learns the mapping from network configuration to predict UE traffic flow distribution $\boldsymbol{\phi}^{\mathrm{c}}$ as follows:

$$\mathcal{M}:\ (\mathcal{A}^{\mathrm{c}},\ \boldsymbol{\theta}^{\mathrm{c}},\ \boldsymbol{q}^{\mathrm{c}}) \longmapsto \boldsymbol{\phi}^{\mathrm{c}}. \tag{27}$$

Since these inputs, in particular $\boldsymbol{\theta}^{\mathrm{c}}$ and potentially $\boldsymbol{q}^{\mathrm{c}}$ through mode choice, are functions of the upgraded cycling configuration $\boldsymbol{w}^{\mathrm{b}}$, we write the surrogate compactly as

$$\mathcal{M}:\ \boldsymbol{w}^{\mathrm{b}} \longmapsto \boldsymbol{\phi}^{\mathrm{c}}. \tag{28}$$

However, since the GCN does not explicitly enforce flow conservation, $\boldsymbol{\phi}^{\mathrm{c}}$ may violate $B\boldsymbol{\phi}^{\mathrm{c}} = \boldsymbol{d}^{\mathrm{c}}$. We restore feasibility by projecting the surrogate prediction onto the flow-conserving subspace via the quadratic program

$$\boldsymbol{\phi}^{c^\star} \;=\; \arg\min_{\boldsymbol{\phi}^{\mathrm{c}} \geq 0} \; \left\| \tilde{\boldsymbol{\phi}}^{\mathrm{c}} - \boldsymbol{\phi}^{\mathrm{c}} \right\|_2^2 \quad \text{s.t.} \quad B\boldsymbol{\phi}^{\mathrm{c}} = \boldsymbol{d}^{\mathrm{c}}. \tag{29}$$

The solution $\boldsymbol{\phi}^{c^\star}$ represents the Euclidean projection of the GCN-predicted flows onto the feasible subspace, ensuring node-level flow conservation and nonnegativity. The corresponding car total system travel time associated with $\boldsymbol{\phi}^{c^\star}$ is denoted as $\mathcal{T}^{\mathrm{c}}(\boldsymbol{w}^{\mathrm{b}})$. Finally, the surrogate-based total system travel time combines the computed bicycle component with the GCN-predicted car component

$$\mathcal{T}_{\mathrm{tot}}(\boldsymbol{w}^{\mathrm{b}}) \;=\; \mathcal{T}_{\mathrm{b}}(\boldsymbol{w}^{\mathrm{b}}) \;+\; \mathcal{T}_{\mathrm{c}}(\boldsymbol{w}^{\mathrm{b}}). \tag{30}$$

This GCN–projection framework provides fast and consistent approximations of network performance, enabling efficient optimization of multi-modal configurations while substantially reducing computational cost. A detailed assessment of the model's predictive accuracy and reliability is presented in the Results section.

F. Optimization Problem (Utilitarian Perspective)

We now introduce the utilitarian formulation of the trade-off problem for upgrading cycling infrastructure. Under a utilitarian perspective, the planner seeks to improve the overall performance of the transport system by minimizing aggregate costs across users. Given a fixed budget B, the planner selects a subset of road segments to upgrade in order to enhance their degree of separation. The objective balances two competing goals: (i) minimizing cyclists' average exposure to motor traffic, and (ii) limiting the aggregate increase in total travel times due to car capacity reallocation. This formulation therefore aims to reduce the overall system cost by jointly accounting for traffic exposure and travel-time inefficiency.

Let the decision variables

$$\boldsymbol{w}^{\mathrm{b}} = \left[w_a^{\mathrm{b}} \right]_{a \in \mathcal{A}^{\mathrm{b}}}^{\top} \tag{31}$$

denote the vector of upgraded DoS values for all bicycle arcs, with the feasible set

$$\Omega = \left\{ \boldsymbol{w}^{\mathrm{b}} \in \{1, \ldots, 6\}^{|\mathcal{A}^{\mathrm{b}}|} : \tilde{w}_a^{\mathrm{b}} \leq w_a^{\mathrm{b}} \leq 6, \; \forall a \in \mathcal{A}^{\mathrm{b}} \right\}. \tag{32}$$

We formulate the problem as a bi-level optimization, where the upper level determines infrastructure upgrades, and the lower level models the resulting cyclist route choices.

At the *upper level* ($\mathbb{U}$), the planner selects upgrades $\boldsymbol{w}^{\mathrm{b}} \in \Omega$ subject to the budget B, minimizing total cyclist exposure and total travel time while controlling the impact on the average car total travel times. The formulation naturally prioritizes the improvement of longer links with both high cyclist and vehicle volumes, since these contribute most to total exposure and travel cost.

At the *lower level* ($\mathbb{L}$), given a specific upgraded configuration $\boldsymbol{w}^{\mathrm{b}}$, cyclist flows are assigned to origin–destination pairs using an all-or-nothing (AON) shortest-path assignment based on the effective cyclist distance defined in Section II-C. This ensures that the induced flow distribution reflects perceived safety across the network.

Problem 2 (Utilitarian Trade-Off Network Upgrade):

$$\mathbb{U}: \quad \boldsymbol{w}^{\mathrm{b}^\star} = \arg\min_{\boldsymbol{w}^{\mathrm{b}} \in \Omega} \; \gamma \cdot \underbrace{\sum_{a \in \mathcal{A}^{\mathrm{b}}} \phi_a^{\mathrm{b}^\star}(\boldsymbol{w}^{\mathrm{b}}) \cdot f_a^{\mathrm{b}}(\boldsymbol{w}^{\mathrm{b}}) \cdot d_a^{\mathrm{b}}}_{\text{Cyclist exposure } J_{\mathcal{E}}(\boldsymbol{w}^{\mathrm{b}})} + (1-\gamma) \cdot \underbrace{\mathcal{T}_{\mathrm{tot}}(\boldsymbol{w}^{\mathrm{b}})}_{\text{Predicted total travel time}} \tag{33}$$

$$\text{s.t.} \quad \sum_{a \in \mathcal{A}^{\mathrm{b}}} d_a^{\mathrm{b}} \cdot \pi(\tilde{w}_a^{\mathrm{b}}, w_a^{\mathrm{b}}) \;\leq\; B, \tag{34}$$

$$\theta_a^{\mathrm{c}}(\boldsymbol{w}^{\mathrm{b}}) = \tilde{\theta}_a^{\mathrm{c}} \left(1 - \frac{z_a}{n_a^{\mathrm{c}}} \right), \tag{35}$$

where the lower-level bicycle flow assignment is

$$\mathbb{L}: \quad \boldsymbol{\phi}^{\mathrm{b}^\star}(\boldsymbol{w}^{\mathrm{b}}) = \arg\min_{\boldsymbol{\phi}^{\mathrm{b}} \in \mathbb{R}_{\geq 0}^{|\mathcal{A}^{\mathrm{b}}|}} \sum_{a \in \mathcal{A}^{\mathrm{b}}} \bar{d}_a^{\mathrm{b}}(\boldsymbol{w}^{\mathrm{b}}) \cdot \phi_a^{\mathrm{b}} \tag{36}$$

$$\text{s.t.} \quad \phi_a^{\mathrm{b}} = \sum_{(u,v) \in \mathcal{V}_{\mathcal{O}}^{\mathrm{b}} \times \mathcal{V}_{\mathcal{D}}^{\mathrm{b}}} q_{(u,v)}^{\mathrm{b}}(\boldsymbol{w}^{\mathrm{b}}) \cdot \mathbb{I}_{a \in p_{(u,v)}^{\mathrm{b}}}, \quad \forall a \in \mathcal{A}^{\mathrm{b}}.$$

The two objective terms are normalized by their values in the original network configuration, so that cyclist exposure and total travel time become dimensionless and comparable. The parameter $\gamma \in [0,1]$ controls the trade-off between the two terms: Larger values of γ give more importance to reducing cyclist exposure, whereas smaller values give more importance to limiting total travel-time impacts. The budget constraint (34) ensures feasibility of the selected infrastructure upgrades, while the capacity relation (35) captures the reduction in car capacity when high-quality separated bicycle infrastructure (DoS $\geq$ 5) is introduced. The lower-level problem (36) assigns bicycle flows through an all-or-nothing shortest-path model under the updated generalized cyclist distances and the new bicycle demand resulting from modal shifts. These flows $\boldsymbol{\phi}^{\mathrm{b}\star}(\boldsymbol{w}^{\mathrm{b}})$ then enter the upper-level objective, ensuring consistency between infrastructure upgrades, route choices, flows, and exposure.

The bi-level optimization in Problem 2 is NP-hard, as it generalizes the 0-1 knapsack problem where each arc upgrade represents an item with cost $\pi(\tilde{w}_a^{\mathrm{b}}, w_a^{\mathrm{b}})$ and benefits both cyclist exposure and traffic flow. The complexity arises from the combinatorial upgrade choices under budget B, and the evaluation of cyclist exposure and non-linear car flow effects from changes in θ_a^{c}.

G. Optimization Problem (Sufficientarian Perspective)

We now adopt a sufficientarian perspective by targeting a minimum acceptable level of exposure-based path protection for as many cycling trips as possible. Empirical evidence shows that perceived cycling safety is strongly shaped by motor-traffic conditions, notably traffic volume and speed, and by the presence and quality of protective infrastructure. When detailed conflict or crash data are not available, these perceptions can be approximated through objective, exposure-based indicators [53], [54], [55]. Following this rationale, we define an exposure-based path sufficiency score that decreases with the average traffic exposure experienced along the selected

cycling path. We use average exposure, rather than cumulative exposure, because the sufficientarian formulation aims to identify paths whose protection level is inadequate relative to their length, rather than penalizing long trips only because they are long. Total cyclist exposure is instead accounted for separately in the utilitarian objective. Therefore, the score below should be interpreted as a normalized indicator of exposure-based path protection, rather than as a direct measure of crash risk or observed safety.

For a bicycle OD pair (u, v) with selected path $p^{\mathrm{b}}_{(u,v)}$, we define

$$S_{p^{\mathrm{b}}_{(u,v)}} \;=\; \frac{1}{\dfrac{\sum_{a\in p^{\mathrm{b}}_{(u,v)}} f^{\mathrm{b}}_a\, d^{\mathrm{b}}_a}{\sum_{a\in p^{\mathrm{b}}_{(u,v)}} d^{\mathrm{b}}_a} \;+\; 1}, \tag{37}$$

where f^{b}_a is the traffic-related exposure factor on arc a, and d^{b}_a is its length. The term 1 keeps the score bounded between 0 and 1. Higher values indicate paths with lower average exposure to motor traffic, while lower values identify paths with weaker exposure-based protection [53], [54], [55].

To determine what constitutes a sufficient level of exposure-based path protection, we adopt a relative threshold approach. Let the global average path sufficiency score be

$$S^{\mathrm{b}}_{\mathcal{G}} = \frac{1}{|\mathcal{V}^{\mathrm{b}}|^2} \sum_{(u,v)\in\mathcal{V}^{\mathrm{b}}\times\mathcal{V}^{\mathrm{b}}} S_{p^{\mathrm{b}}_{(u,v)}}. \tag{38}$$

We then define a relative path-sufficiency threshold S_{thr} as a proportion of the average score $S^{\mathrm{b}}_{\mathcal{G}}$, using a parameter λ that takes values from the set $\{10\%, 20\%, 30\%, 40\%, 50\%\}$:

$$S_{\text{thr}} = \lambda S^{\mathrm{b}}_{\mathcal{G}}. \tag{39}$$

This method follows the sufficientarian logic used in transport justice [6], [29], where sufficiency thresholds are defined as a proportion of an average or reference value. Cycling paths whose scores fall below this threshold, i.e., $S_{p^{\mathrm{b}}_{(u,v)}} \le S_{\text{thr}}$, are considered insufficiently protected in exposure-based terms.

To quantify this shortfall, and following a similar approach to [30], we adopt a linear form of the Foster–Greer–Thorbecke (FGT) index, originally developed for poverty measurement, which is well suited to capture transport injustice [6], [27]. In the FGT framework, deprivation is measured as the normalized gap between an observed value and a predefined threshold. Analogously, here we measure the exposure-based path insufficiency as the normalized deficit of the path score relative to the sufficiency threshold

$$\mathcal{I}_{p^{\mathrm{b}}_{(u,v)}} = \max\left\{0,\; \frac{S_{\text{thr}} - S_{p^{\mathrm{b}}_{(u,v)}}}{S_{\text{thr}}}\right\}. \tag{40}$$

This quantity is zero when the selected path meets or exceeds the threshold and increases continuously as the path-sufficiency score falls further below it. It should therefore be interpreted as a normalized exposure-based protection deficit, rather than as a direct measure of crash risk, welfare loss, or accessibility deprivation. This deficit measure enables aggregation across trips and regions to characterize exposure-based path insufficiency at the network level.

Dividing the network into regions $\mathcal{R} = \{1, \ldots, R\}$, where $\mathcal{V}^{\mathrm{b}}_r \subseteq \mathcal{V}^{\mathrm{b}}$ represents the set of bicycle nodes in region r, allows the exposure-based path insufficiency to be evaluated spatially. We use a demand-weighted average over all bicycle trips originating in region r,

$$\mathcal{I}_r(\boldsymbol{w}^{\mathrm{b}}) = \frac{\sum_{(u,v):\, u\in\mathcal{V}^{\mathrm{b}}_r} \mathcal{I}_{p^{\mathrm{b}}_{(u,v)}} \cdot q^{\mathrm{b}}_{(u,v)}}{\sum_{(u,v):\, u\in\mathcal{V}^{\mathrm{b}}_r} q^{\mathrm{b}}_{(u,v)}}. \tag{41}$$

Here, $\mathcal{I}_{p^{\mathrm{b}}_{(u,v)}}$ denotes the exposure-based path insufficiency of the selected cycling path, and $q^{\mathrm{b}}_{(u,v)}$ is the bicycle demand for OD pair (u, v). The total population-weighted exposure-based path insufficiency across the network is given by

$$J_{\mathrm{SI}}(\boldsymbol{w}^{\mathrm{b}}) = \frac{\sum_{r\in\mathcal{R}} n^{\mathrm{b}}_r\, \mathcal{I}_r(\boldsymbol{w}^{\mathrm{b}})}{\sum_{r\in\mathcal{R}} n^{\mathrm{b}}_r}, \tag{42}$$

where n^{b}_r represents the number of cyclists in region r.

To formulate the sufficientarian perspective in terms of trip travel time for both cars and bicycles, we first introduce a path travel-time threshold denoted by T_{thr}; for example, $T_{\text{thr}} = 20$ min $= 1200$ s, as used in [30]. For bicycles, given any trip (u, v) and its corresponding chosen path $p^{\mathrm{b}}_{(u,v)}$, and using the previously defined path travel time $T^{\mathrm{b}}_{p_{(u,v)}}$ in (12), the bicycle trip insufficiency is defined as

$$\mathcal{I}^{\mathrm{b}}_{p_{(u,v)}} \;=\; \max\left\{0,\; \frac{T^{\mathrm{b}}_{p_{(u,v)}} - T_{\text{thr}}}{T_{\text{thr}}}\right\}. \tag{43}$$

This quantity measures the relative excess travel time above the acceptable threshold T_{thr}. A trip between u and v is considered trip-sufficient if its travel time does not exceed the threshold, i.e., $T^{\mathrm{b}}_{p_{(u,v)}} \le T_{\text{thr}}$, and trip-insufficient otherwise. Hence, $\mathcal{I}^{\mathrm{b}}_{p_{(u,v)}}$ represents the relative excess travel time beyond this limit.

We use the same demand-weighted regional aggregation for trip-time insufficiency, but now the trip-level quantity is $\mathcal{I}^{\mathrm{b}}_{p_{(u,v)}}$, which measures excess bicycle travel time relative to T_{thr}. Thus, cyclists' regional trip insufficiency is defined as

$$\mathcal{I}^{\mathrm{b}}_r(\boldsymbol{w}^{\mathrm{b}}) \;=\; \frac{\sum_{(u,v):\, u\in\mathcal{V}^{\mathrm{b}}_r} \mathcal{I}^{\mathrm{b}}_{p_{(u,v)}} \cdot q^{\mathrm{b}}_{(u,v)}}{\sum_{(u,v):\, u\in\mathcal{V}^{\mathrm{b}}_r} q^{\mathrm{b}}_{(u,v)}}. \tag{44}$$

This formulation ensures that regions with higher cycling demand exert a proportionally greater influence on the regional insufficiency measure. To evaluate cyclist trip insufficiency at the network level, we define the cyclist-weighted network trip insufficiency as

$$J^{\mathrm{b}}_{\mathrm{TI}}(\boldsymbol{w}^{\mathrm{b}}) \;=\; \frac{\sum_{r\in\mathcal{R}} n^{\mathrm{b}}_r \cdot \mathcal{I}^{\mathrm{b}}_r(\boldsymbol{w}^{\mathrm{b}})}{\sum_{r\in\mathcal{R}} n^{\mathrm{b}}_r}. \tag{45}$$

This indicator provides a population-weighted measure of bicycle trip insufficiency, reflecting the severity of travel-time exceedances across regions. Analogously, we define the driver-weighted network trip insufficiency $J^{\mathrm{c}}_{\mathrm{TI}}(\boldsymbol{w}^{\mathrm{b}})$ for car users.

Finally, we define the sufficientarian bi-level optimization problem, where $\mathbb{U}_{\text{thr}}$ seeks to reduce exposure-based path insufficiency for cyclists and trip insufficiency for both cyclists and drivers. As in the utilitarian formulation, the lower level

$\mathbb{L}$ determines the resulting cyclist flow distribution under the upgraded network.

Problem 3 (Sufficientarian Trade-Off Network-Upgrade):

$$\mathbb{U}_{\text{thr}}: \quad \boldsymbol{w}^{\text{b}^\star} = \arg\min_{\boldsymbol{w}^{\text{b}} \in \Omega} \; \gamma \cdot \underbrace{J_{\text{SI}}(\boldsymbol{w}^{\text{b}})}_{\text{Exposure-based path insufficiency}} + (1-\gamma) \cdot \eta \cdot \underbrace{\left[J^{\text{b}}_{\text{TI}}(\boldsymbol{w}^{\text{b}}) + J^{\text{c}}_{\text{TI}}(\boldsymbol{w}^{\text{b}})\right]}_{\text{Trip insufficiency}} \tag{46}$$

$$\text{s.t.} \quad \sum_{a \in \mathcal{A}^{\text{b}}} d^{\text{b}}_a \cdot \pi(\tilde{w}^{\text{b}}_a, w^{\text{b}}_a) \;\le\; B, \tag{47}$$

$$\theta^{\text{c}}_a(\boldsymbol{w}^{\text{b}}) = \tilde{\theta}^{\text{c}}_a \cdot \left(1 - \frac{z_a}{n^{\text{c}}_a}\right), \tag{48}$$

and as before, $\mathbb{L}$ models the resulting flow distribution.

$$\mathbb{L}: \quad \boldsymbol{\phi}^{\text{b}^\star}(\boldsymbol{w}^{\text{b}}) = \arg\min_{\boldsymbol{\phi}^{\text{b}} \in \mathbb{R}^{|\mathcal{A}^{\text{b}}|}_{\ge 0}} \sum_{a \in \mathcal{A}^{\text{b}}} \bar{d}^{\text{b}}_a(\boldsymbol{w}^{\text{b}}) \cdot \phi^{\text{b}}_a \tag{49}$$

$$\text{s.t.} \quad \phi^{\text{b}}_a = \sum_{(u,v) \in \mathcal{V}^{\text{b}}_{\mathcal{O}} \times \mathcal{V}^{\text{b}}_{\mathcal{D}}} q^{\text{b}}_{(u,v)}(\boldsymbol{w}^{\text{b}}) \cdot \mathbb{I}_{a \in p^{\text{b}}_{(u,v)}}, \quad \forall a \in \mathcal{A}^{\text{b}}.$$

A summary of the model notation is given in Table II.

Symbol	Description	Domain	Unit
$q^{\text{b/c}}_{(u,v)}$	Bike/Car demand from u to v	$\mathbb{R}_{\ge 0}$	bike/day
$\bar{d}^{\text{b}}_{(u,v)}$	Cyclist distance u-v	$\mathbb{R}_{\ge 0}$	m
$F(a)$	Annual average daily traffic adjacent to $a \in \mathcal{A}^{\text{b}}$	$\mathbb{R}_{\ge 0}$	veh/day
S_{thr}	Safety path threshold	$\mathbb{R}_{\ge 0}$	-
T_{thr}	Time path threshold	$\mathbb{R}_{\ge 0}$	-
$\tilde{\boldsymbol{w}}^{\text{b}}$	Vector of original DoS for all $a \in \mathcal{A}^{\text{b}}$	$\{1,\dots,6\}$	-
$\boldsymbol{w}^{\text{b}}$	Candidate vector DoS for all $a \in \mathcal{A}^{\text{b}}$	$\{1,\dots,6\}$	-
$\phi^{\text{b}}_a(\boldsymbol{w}^{\text{b}})$	Bicycle flow on arc $a \in \mathcal{A}^{\text{b}}$	$\mathbb{R}_{\ge 0}$	bike/day
$\phi^{\text{c}}_a(\boldsymbol{w}^{\text{b}})$	Car flow on arc $a \in \mathcal{A}^{\text{c}}$	$\mathbb{R}_{\ge 0}$	car/day
$\bar{D}^{\text{b}}_a(\boldsymbol{w}^{\text{b}})$	Average cyclist distance	$\mathbb{R}_{\ge 0}$	m
$\mathcal{E}_a(\boldsymbol{w}^{\text{b}})$	Cyclist exposure on $a \in \mathcal{A}^{\text{b}}$	$\mathbb{R}_{\ge 0}$	-
$\theta^{\text{c}}_a(\boldsymbol{w}^{\text{b}})$	Updated car capacity on $a \in \mathcal{A}^{\text{c}}$	$\mathbb{R}_{\ge 0}$	veh/day
$\pi(\tilde{w}^{\text{b}}_a, w^{\text{b}}_a)$	Upgrade cost on $a \in \mathcal{A}^{\text{b}}$	$\mathbb{R}_{\ge 0}$	€
$\mathcal{T}_{\text{tot}}(\boldsymbol{w}^{\text{b}})$	Predicted total system travel time	$\mathbb{R}_{\ge 0}$	$\frac{\text{travelers}\cdot\text{hr}}{\text{day}}$
$J_{\mathcal{E}}(\boldsymbol{w}^{\text{b}})$	Total cyclist exposure	$\mathbb{R}_{\ge 0}$	-
$J_{\text{SI}}(\boldsymbol{w}^{\text{b}})$	Exposure-based path insufficiency	$\mathbb{R}_{\ge 0}$	-
$J^{\text{b}}_{\text{TI}}(\boldsymbol{w}^{\text{b}})$	Total bike-time insufficiency	$\mathbb{R}_{\ge 0}$	-
$J^{\text{c}}_{\text{TI}}(\boldsymbol{w}^{\text{b}})$	Total car-time insufficiency	$\mathbb{R}_{\ge 0}$	-

TABLE II: Abbreviated notation for variables in trade-off network model. Notation for optimization variables depending on the upgrade configuration $\boldsymbol{w}^{\text{b}}$

III. Solution Algorithm

To address the computational complexity of the proposed bi-level optimization problems within a practical time frame, we adopt a genetic algorithm (GA) as the main solution approach. The GA is applied to both problem formulations—the utilitarian problem (Problem 2) and the sufficientarian problem (Problem 3). To improve convergence and reduce search time, the initial GA population is warm-started using a connectivity-based heuristic inspired by the Maximize Largest Connected Component (MLCC) approach [14]. This heuristic is used solely to generate an initial feasible solution that enhances the connectivity of high-safety (DoS 6) cycling infrastructure, particularly in areas exhibiting higher trip insufficiency. As shown in [37], such a warm-start strategy can significantly improve GA performance in cycling network optimization. No separate MLCC-based optimization results are reported.

A. Genetic Algorithm with Efficient Budget Spending (Utilitarian / Sufficientarian)

The GA iteratively explores upgrade configurations $\tilde{\boldsymbol{w}}^{\text{b}}$ under a fixed budget B. Each candidate solution represents a vector of upgraded DoS values for the bicycle network and is evaluated using the corresponding objective function, in the utilitarian formulation minimizes total cyclist exposure $J_{\mathcal{E}}(\tilde{\boldsymbol{w}}^{\text{b}})$ (utilitarian perspective), on the other hand, the sufficientarian formulation minimizes safety or time insufficiency $J_{\text{SI}}(\tilde{\boldsymbol{w}}^{\text{b}})$ or $J^{\text{b}}_{\text{TI}}(\tilde{\boldsymbol{w}}^{\text{b}})$ (sufficientarian perspective). An initial population of upgrade solutions is generated randomly within the feasible set Ω (see Section II-G). Solutions are evolved through selection, crossover, and mutation operations. Infeasible individuals exceeding the available budget are discarded. A budget optimization phase ensures efficient spending: Whenever residual budget $\tilde{B}$ remains, the algorithm greedily upgrades arcs with the lowest DoS until no feasible upgrades are possible. This guarantees near-complete budget use without compromising solution feasibility. While the GA does not guarantee global optimality, it provides high-quality, computationally tractable solutions for both formulations. The pseudocode below summarizes the GA procedure, common to both variants.

- $\tilde{B}$: Remaining budget after the initial solution $\tilde{\boldsymbol{w}}^{\text{b}}$ has been evaluated.
- Δ_a: Cost to upgrade arc a to DoS $= 6$, i.e., $\pi(\tilde{w}^{\text{b}}_a, 6)$.

Algorithm 1 Genetic Algorithm

Input: Candidate weights $\tilde{\boldsymbol{w}}^{\text{b}}$, remaining budget $\tilde{B}$
Output: Updated weights $\tilde{w}^{\text{b}}_a$, remaining budget $\tilde{B}$
for $a \in \mathcal{A}^{\text{b}}$ sorted by DoS(a) **do**
  Compute upgrade cost $\Delta_a = \pi(\tilde{w}^{\text{b}}_a, w_{\text{DoS}_6})$
  if $\tilde{B} \ge \Delta_a$ **then**
    Upgrade arc $\tilde{w}^{\text{b}}_{a_\sim}$ to $w_{\text{DoS}} = 6$
    Update budget $\tilde{B} \leftarrow \tilde{B} - \Delta_a$
  end if
  if $\tilde{B} < 0$ **then**
    Exit
  end if
end for
Return: Updated $\tilde{w}^{\text{b}}_a$, remaining $\tilde{B}$

Now that we have presented the solution algorithms, we will introduce the case study of this work.

IV. Grenoble Cycling Network Case Study

This section introduces the parameters for the AON flow assignment, a sensitivity analysis for budget selection, and a case study of Grenoble, France. It also examines the results obtained using the MLCC, the utilitarian and sufficientarian formulations in comparison to the original scenario.

A. Parameters and Algorithm Comparisons

To solve the AON traffic assignment problem and determine cyclist flows, as described in Section II-C, we use parameters

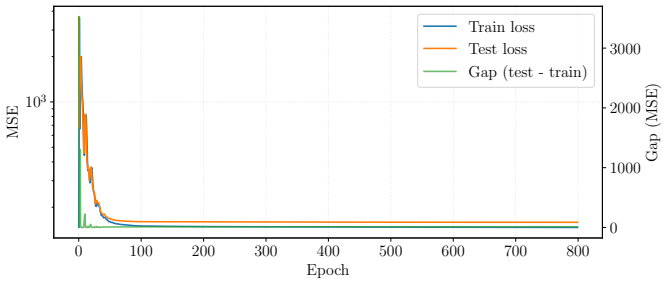
that are better suited for the model, as suggested by [38]. Specifically, we set $\rho = 0.04$, $\kappa = 0.14$, and $\psi = 0.84$. The slope factor s_a is determined in Table III, where downhill routes are considered the reciprocal of the slope factor when traversing in the opposite direction. In the experiments, the

Slope (m_a) %	Uphill Factor (s_a)	Downhill Factor (s_a)
$m_a < 2$	1.0	1.0
$2 \leq m_a < 4$	1.371	$1/1.371 \approx 0.729$
$4 \leq m_a < 6$	2.203	$1/2.203 \approx 0.454$
$m_a \geq 6$	4.239	$1/4.239 \approx 0.236$

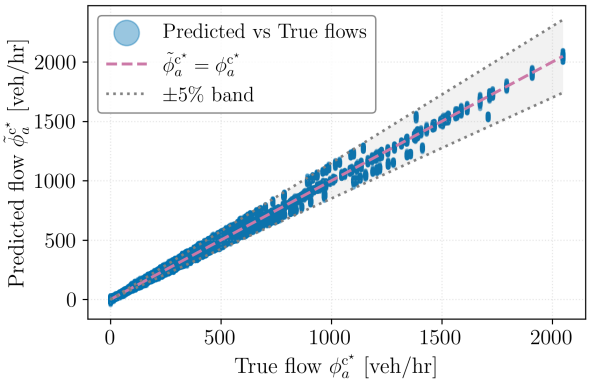
TABLE III: Slope factors for cycling distance calculation as in [39], [38].

lane width threshold is set to $l_a \leq 4.5\,\mathrm{m}$, corresponding to the standard lane width recommended in France [56]. Using the historical data from Fig. 4, the estimated parameters of the bicycle mode-choice model are $\alpha = 0.128$ and $\beta = -0.886$, in this logit formulation, α is an alternative specific constant that captures unobserved factors favoring cycling over car use such as cultural preferences, infrastructure quality, or policy incentives when the perceived cycling distance is zero. The parameter $\beta < 0$ quantifies the sensitivity of cycling adoption to the average cycling distance: as distances increase, the perceived disutility of cycling rises, leading to a lower probability of choosing the bicycle mode. The estimated values indicate a moderate baseline preference for cycling and a strong negative influence of distance on mode choice, consistent with observed behavioral patterns in the Grenoble metropolitan area. The 912 UE samples were generated in parallel, requiring a total of 5,928 CPU hours. Fig. 5 **(a)**–**(b)** illustrate the evolution of the training, testing, and generalization gap losses over 800 epochs, obtained using PyTorch [57] to train the GCN. They also show how the predicted flow progressively converges toward the optimal car flow distribution derived from the UE model. To substantiate the approximation quality, we report standard regression error metrics between predicted and UE flows. The mean absolute error (MAE) is 68.9 veh/h and the root mean squared error (RMSE) is 105.9 veh/h. The coefficient of determination is $R^2 = 0.9696$, indicating that the surrogate explains most of the variance in UE flows. In addition, 51.5% of link flows are predicted within $\pm 5\%$ of the UE value, and 82.1% within $\pm 10\%$. Finally, the 95th and 99th percentiles of the absolute error are 227 and 369 veh/h, respectively, showing that the largest discrepancies occur on a small fraction of high-flow links. These errors are relevant because car travel-time estimates influence the upper-level trade-off between cycling improvements and driver trip insufficiency. Therefore, the GCN should be interpreted as a computationally efficient surrogate for guiding the search, rather than as an exact replacement for UE assignment. While the reported accuracy supports its use for identifying broad performance trends and promising infrastructure configurations, small differences between near-optimal solutions should be interpreted with caution. These results also motivate the feasibility-projection step introduced in Section II-E1.

Let $B_{\max}$ represent the budget required to upgrade the entire network to DoS $= 6$. The cost per meter $\pi(w_a^{\mathrm{b}}, \tilde{w}_a^{\mathrm{b}})$ of upgrading arc a from w_a^{b} to $\tilde{w}_a^{\mathrm{b}}$ is given by the matrix below. The total cost of upgrading an arc is computed as $d_a^{\mathrm{b}} \cdot \pi(w_a^{\mathrm{b}}, \tilde{w}_a^{\mathrm{b}})$, where



(a) Training, test and gap progress.



(b) Predicted vs true flows using the surrogate model

Fig. 5: On the top we present the epochs evolution with the 912 samples created for solving Problem 1 and on the bottom how the predicted flows approaches the true flow found by the UE.

d_a^{b} is the arc length, making the cost distance-dependent. The cost matrix Π is adapted from Cerema's reported unit construction costs for cycling infrastructure upgrades in France [58].

$$\Pi = \begin{array}{c|cccccc} \tilde{w}_a \backslash w_a & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & 0€ & 150€ & 250€ & 300€ & 500€ & 750€ \\ 2 & 0€ & 0€ & 100€ & 150€ & 350€ & 600€ \\ 3 & 0€ & 0€ & 0€ & 50€ & 250€ & 500€ \\ 4 & 0€ & 0€ & 0€ & 0€ & 200€ & 450€ \\ 5 & 0€ & 0€ & 0€ & 0€ & 0€ & 250€ \\ 6 & 0€ & 0€ & 0€ & 0€ & 0€ & 0€ \end{array} \tag{50}$$

According to the Grenoble cycling investment plan (March 2025) [59], the planned budget is about 24.5M €, which corresponds to $B \approx 0.15\, B_{\max}$. We therefore solve Problem 2 (utilitarian) and Problem 3 (sufficientarian) under this budget constraint.

For each formulation, we explore the trade-off between the competing objectives by solving the corresponding bi-level problem for a discrete set of weights

$$\Lambda = \{0, 10\%, 20\%, \ldots, 100\%\},$$

where $\gamma \in \Lambda$ controls the balance between the safety-related term and the travel-time/accessibility-related term. This yields, for each γ, one candidate solution and one point on the corresponding Pareto front (reported in Fig. 6).

In the sufficientarian formulation, we additionally test several safety-threshold levels through

$$\lambda \in \{10\%, 20\%, 30\%, 40\%, 50\%\},$$

which define increasingly demanding path-safety requirements. For each λ, the algorithm is run for all $\gamma \in \Lambda$, producing a family of trade-off curves.

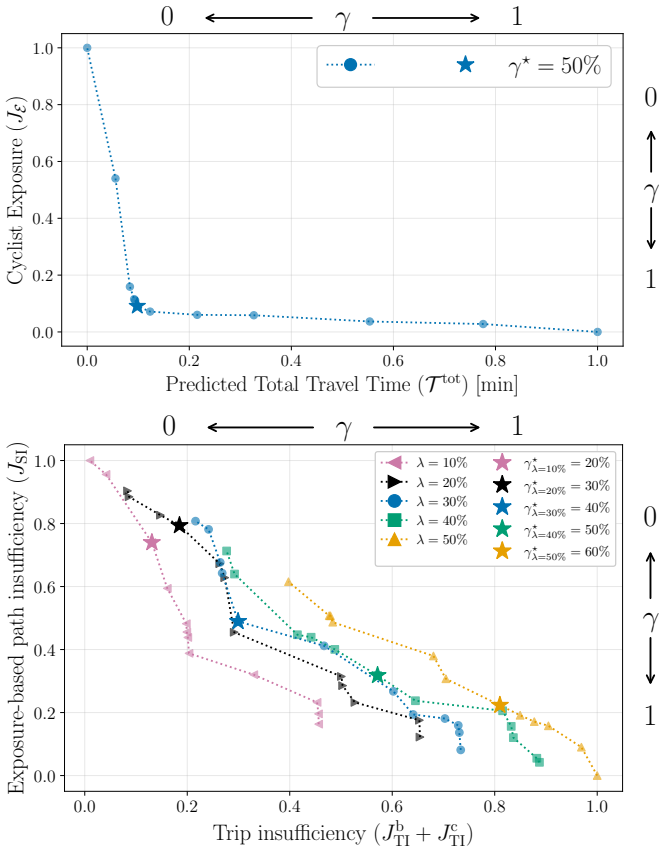
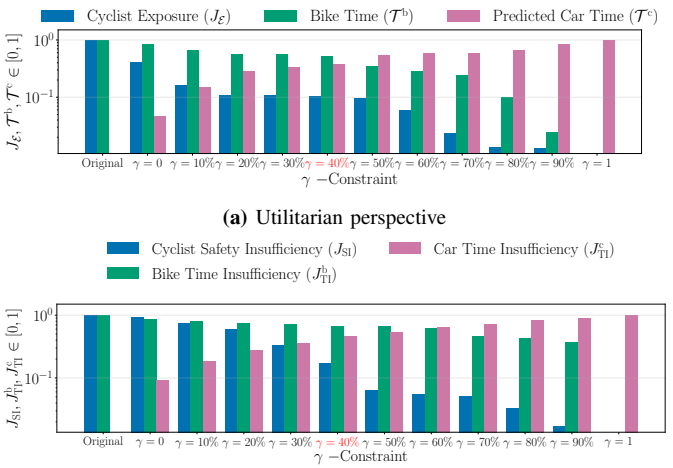


Fig. 6: Normalized travel time versus cyclist exposure. The top panel shows the Pareto fronts obtained from the trade-off Problem 2 under varying values of $\epsilon \in \Lambda$ and $\alpha \in \Phi$. The bottom panel presents the corresponding results for the sufficientarian formulation (Problem 3).

Because the two objectives are expressed in different units, we do not define an "optimal" compromise point (e.g., via Euclidean distance to the origin). Instead, selecting γ is ultimately a policy decision that reflects how decision makers value safety improvements relative to travel-time impacts. To support interpretation and visualization of the results, we highlight one representative weight $\gamma^\star_\lambda$ for each threshold level λ, chosen solely for illustration: $\gamma^\star_{10\%} = 20\%$, $\gamma^\star_{20\%} = 30\%$, $\gamma^\star_{30\%} = 40\%$, $\gamma^\star_{40\%} = 50\%$, $\gamma^\star_{50\%} = 60\%$. These $\gamma^\star_\lambda$ values are used only to exemplify the behavior of the utilitarian and sufficientarian solutions in subsequent figures and discussion; they should not be interpreted as mathematically optimal trade-offs.

Fig. 6 compares the sets of solutions produced by the utilitarian formulation (Problem 2) and the sufficientarian formulation (Problem 3). As expected, small γ values emphasize minimizing travel-time/accessibility impacts, while large γ values emphasize cyclist safety objectives, typically at the cost of higher travel-time impacts. Lower safety-threshold levels (smaller λ) generally allow larger safety gains with smaller impacts on car travel times, whereas higher λ values impose stricter requirements and may lead to larger disruptions in the motorized network.



(a) Utilitarian perspective

(b) Sufficientarian perspective $\lambda = 40\%$

Fig. 7: Normalized cyclist exposure, travel time, safety path insufficiency, cyclist and drivers trip insufficiency trade-off for $\lambda = 40\%$.

B. Algorithms Comparison

To illustrate these effects, Fig. 8 reports the distribution of path-safety scores and the corresponding number of cyclists for each method. Overall, the sufficientarian formulation yields a larger share of cyclists traveling on routes that exceed the exposure-based path-protection threshold. Fig. 9 extends this analysis to both cyclists and drivers by showing the proportions of users above and below the trip-sufficiency threshold. While the utilitarian formulation reduces cyclist trip insufficiency, only the sufficientarian formulation keeps a larger share of drivers below the time threshold, thereby limiting travel-time exceedances despite the road-capacity reductions induced by bike-lane upgrades. Fig. 10 further summarizes, for $\lambda = 40\%$, the percentage of cyclists above and below the exposure-based path-protection threshold. In this setting, the sufficientarian formulation improves exposure-based path-sufficiency score while also maintaining comparatively low increases in car travel times, resulting in the smallest impact on motorized traffic. Finally, Fig. 11 compares the DoS distribution and the spatial pattern of regional exposure-based path insufficiency before and after solving the utilitarian and sufficientarian formulations ($\lambda = 40\%$). Whereas the utilitarian formulation tends to distribute investments more evenly, the sufficientarian formulation achieves the largest reductions in regional exposure-based path insufficiency and bicycle trip insufficiency, while also better limiting the negative trip-time impacts on drivers. Fig. 12 presents the sufficientarian outcomes of Problem 2 by reporting the upper-level objective $\mathbb{U}_{\text{thr}}$ across candidate solutions and threshold values λ. When λ is small, exposure-based path insufficiency remains low; as λ increases, insufficiency rises. Across thresholds, the sufficientarian formulation consistently achieves the largest reductions in exposure-based path insufficiency.

C. Total System Performance

To better interpret these findings, we compare them to a reference scenario in which safety-path insufficiency is not

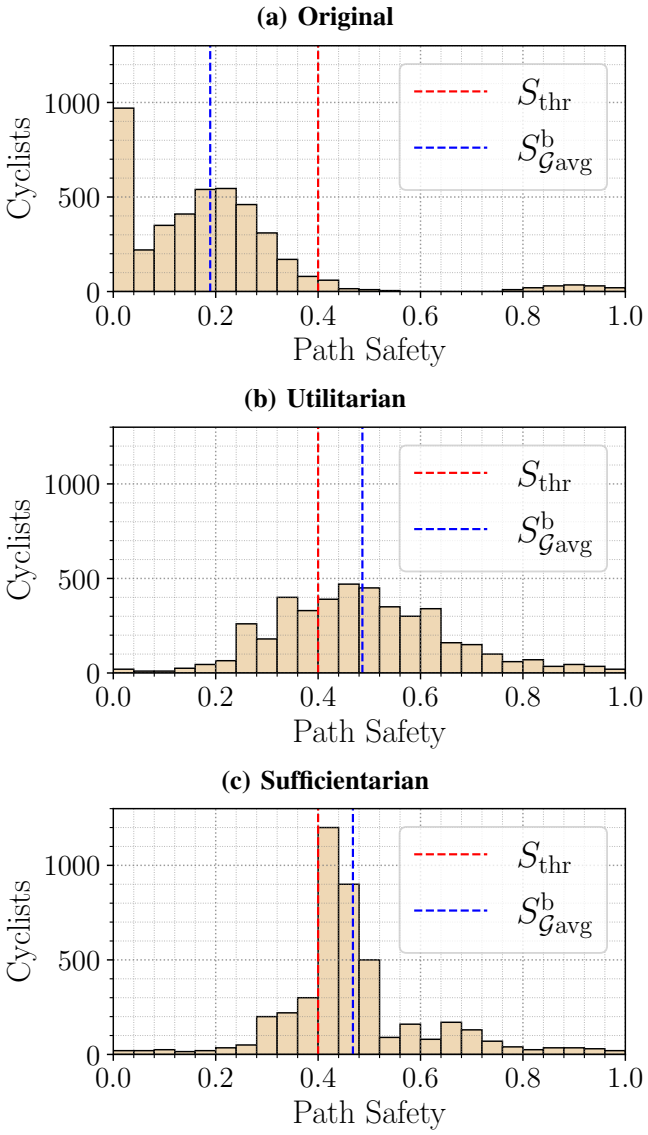


Fig. 8: Exposure-based path-sufficiency score versus the number of cyclists above and below the threshold $\lambda = 40\%$ under a budget of B. The dashed red and blue lines indicate the path-sufficiency threshold s_{thr} and the average exposure-based path-sufficiency score $S^{\text{b}}_{\mathcal{G}_{\text{avg}}}$, respectively.

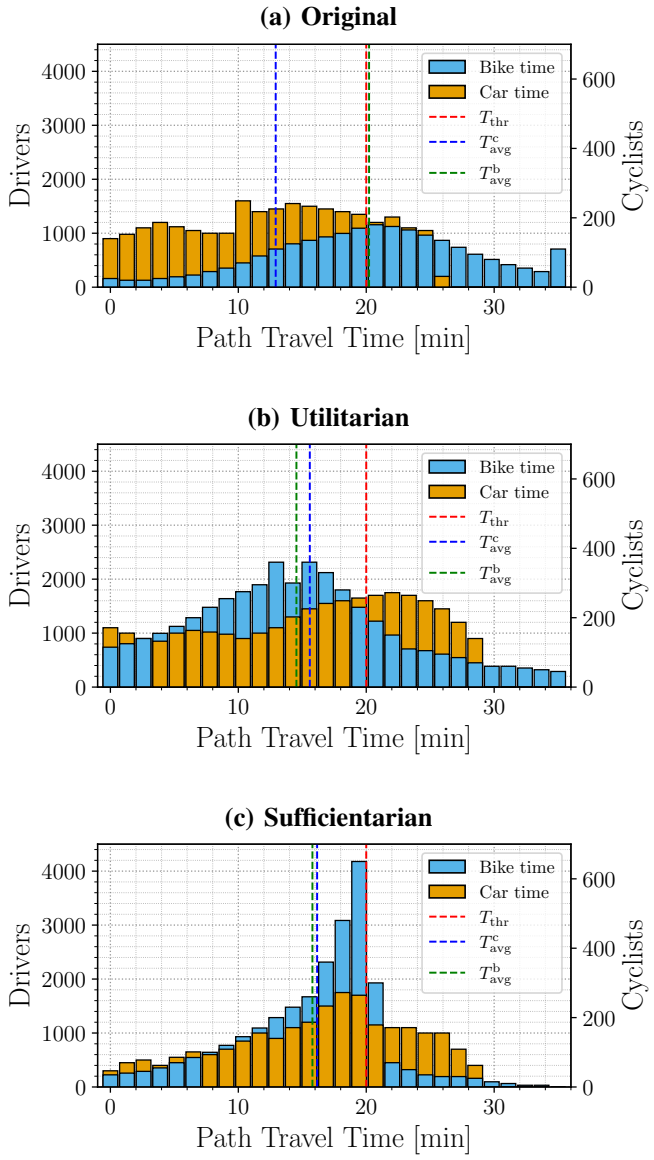


Fig. 9: Drivers (left y-axis) and cyclists (right y-axis) trip insufficiency versus the number of individuals above and below the threshold $\lambda = 40\%$ under a budget of $B^*_\lambda \approx 0.15 \cdot B_{\max}$ for the different optimization algorithms. The dotted red, blue, and green lines indicate the trip-insufficiency threshold T_{thr}, the average bicycle trip time $T^{\text{b}}_{\mathcal{G}_{\text{avg}}}$, and the average car trip time $T^{\text{c}}_{\mathcal{G}_{\text{avg}}}$ of the entire network, respectively.

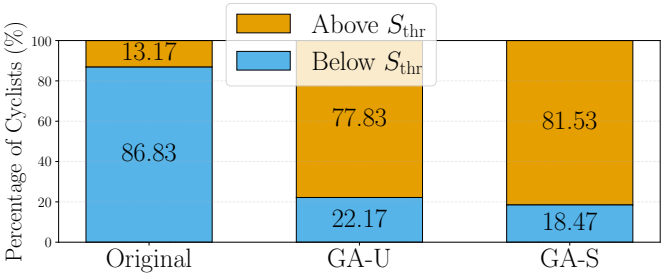
explicitly considered, namely Problem 2, which focuses solely on minimizing cyclists' exposure to motor traffic. As shown in the top panel of Fig. 13, the utilitarian formulation achieves the largest overall reduction in cyclist exposure. Nevertheless, the sufficientarian formulation attains comparable exposure reductions across the evaluated thresholds, despite being designed to balance safety-path insufficiency and exposure.

To assess whether the proposed framework also improves the overall efficiency of the multi-modal system, we evaluate the total system travel time defined in Eq. (30),

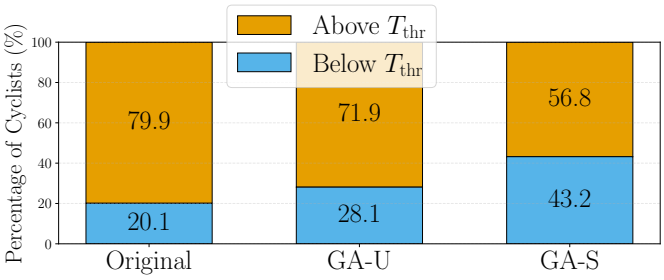
$$\mathcal{T}_{\text{tot}}(\boldsymbol{w}^{\text{b}}) = \mathcal{T}_{\text{b}}(\boldsymbol{w}^{\text{b}}) + \mathcal{T}_{\text{c}}(\boldsymbol{w}^{\text{b}}). \tag{51}$$

In the results reported below, $\mathcal{T}_{\text{tot}}$ is a demand-weighted time metric: the bicycle component aggregates OD travel times weighted by bicycle demand, while the car component aggregates link/path travel times weighted by car flows (or OD demand when using the surrogate predictor). The accompanying modal-share curves report how trips are reallocated across modes through the mode-shift model.

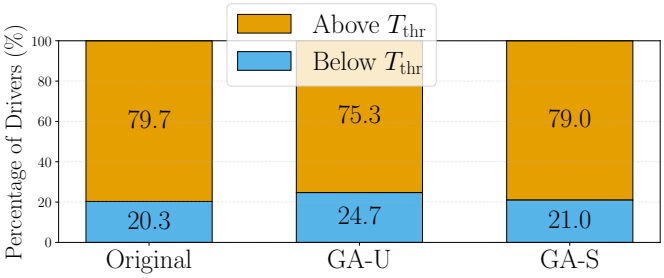
Fig. 14 shows that, in the Grenoble case study, $\mathcal{T}_{\text{tot}}(\boldsymbol{w}^{\text{b}})$ decreases for all optimized configurations relative to the baseline (from 463.7 to 437.5 and 432.8 thousand hours), consistent with a shift away from cars and the resulting reduction in car travel time. Among the evaluated methods, the sufficientarian perspective yields the largest reduction in $\mathcal{T}_{\text{tot}}$, suggesting that a sufficientarian perspective can improve safety outcomes while maintaining (and here improving) overall system efficiency.



(a) Exposure-based path insufficiency $\lambda = 40\%$



(b) Cyclist trip insufficiency $\lambda = 40\%$



(c) Drivers trip insufficiency $\lambda = 40\%$

Fig. 10: Normalized exposure-based path insufficiency, bike path-time insufficiency and car path-time insufficiency for $\lambda = 40\%$.

V. Conclusion and Future Work

This work introduced a multi-modal bi-level optimization framework that combines transport justice, modal-shift modeling, and traffic assignment for cycling infrastructure planning. Applied to the Grenoble network, the results show that cycling infrastructure upgrades can reduce cyclist exposure and exposure-based path insufficiency while keeping increases in total system travel time limited for the tested budget and parameter settings. The comparison between utilitarian and sufficientarian formulations illustrates how different justice principles lead to different infrastructure priorities: the utilitarian formulation emphasizes aggregate exposure and travel-time performance, whereas the sufficientarian formulation more directly targets users experiencing inadequate exposure-based path protection or excessive trip travel times. Among the tested configurations, the sufficientarian genetic algorithm provides the most balanced outcome, reducing exposure-based path and trip insufficiency while maintaining competitive overall performance. However, these conclusions are based on selected values of the trade-off and sufficiency parameters, chosen to illustrate representative outcomes; future work will extend the mode-shift component by integrating additional social and behavioral determinants beyond infrastructure (e.g., city investments, policy incentives, and historical context) to better reflect real-world travel behavior [51]. The framework will also be validated across several cities to assess robustness across different network structures and to strengthen its policy relevance.

Acknowledgment

We thank Dr. I. New for proofreading this paper and for re-humanizing sections where LLMs had been used to improve grammar and flow. All resulting text was reviewed and revised by the authors.

References

[1] Jan Garrard, Chris Rissel, and Adrian Bauman. Health benefits of cycling. *City cycling*, 31:31–56, 2012.

[2] Thomas Götschi, Jan Garrard, and Billie Giles-Corti. Cycling as a part of daily life: a review of health perspectives. *Transport Reviews*, 36(1):45–71, 2016.

[3] Elliot Fishman, Simon Washington, and Narelle Haworth. Barriers and facilitators to public bicycle scheme use: A qualitative approach. *Transportation research part F: traffic psychology and behaviour*, 15(6):686–698, 2012.

[4] Brendan M Lawrence, Jennifer A Oxley, David B Logan, and Mark R Stevenson. Cyclist exposure to the risk of car door collisions in mixed function activity centers: A study in melbourne, australia. *Traffic injury prevention*, 19(sup1):S164–S168, 2018.

[5] John Pucher, Ralph Buehler, Dafna Merom, and Adrian Bauman. Walking and cycling in the united states, 2001–2009: evidence from the national household travel surveys. *American journal of public health*, 101(S1):S310–S317, 2011.

[6] Karel Martens. *Transport Justice: Designing Fair Transportation Systems*. Taylor & Francis, 2017.

[7] Mahmoud Mesbah, Russell Thompson, and Sara Moridpour. Bilevel optimization approach to design of network of bike lanes. *Transportation research record*, 2284(1):21–28, 2012.

[8] Heather Lauren Smith and Ali Haghani. A mathematical optimization model for a bicycle network design considering bicycle level of service. Technical report, University of Maryland, College Park, 2012.

[9] Jennifer Duthie and Avinash Unnikrishnan. Optimization framework for bicycle network design. *Journal of Transportation engineering*, 140(7):04014028, 2014.

[10] Mads Paulsen and Jeppe Rich. Societally optimal expansion of bicycle networks. *Transportation Research Part B: Methodological*, 174:102778, 2023.

[11] CS Shui and WL Chan. Optimization of a bikeway network with selective nodes. *Sustainability*, 11(23):6531, 2019.

[12] Pietro Folco, Laetitia Gauvin, Michele Tizzoni, and Michael Szell. Data-driven micromobility network planning for demand and safety. *Environment and planning B: Urban analytics and city science*, 50(8):2087–2102, 2023.

[13] Stefano Ardizzoni, Mattia Laurini, Rafael Praxedes, Luca Consolini, and Marco Locatelli. Identification of cyclists' route choice criteria. In *2024 IEEE 63rd Conference on Decision and Control (CDC)*, pages 6975–6980. IEEE, 2024.

[14] Manuel Campero Jurado, Carlos Canudas de Wit, Giovanni de Nunzio, and Rémi Mourgues. A general graph-based optimization framework for improving safety and comfort in cycling networks. In *27th IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 2024.

[15] Nina Wiedemann, Christian Nöbel, Lukas Ballo, Henry Martin, and Martin Raubal. Bike network planning in limited urban space. *Transportation Research Part B: Methodological*, 192:103135, 2025.

[16] Hussein Mahfouz, Robin Lovelace, and Elsa Arcaute. A road segment prioritization approach for cycling infrastructure. *Journal of transport geography*, 113:103715, 2023.

[17] Peter Cox. Theorising infrastructure: a politics of spaces and edges. In *The Politics of Cycling Infrastructure*, pages 15–34. Policy Press, 2020.

[18] Ralph Buehler and Jennifer Dill. Bikeway networks: a review of effects on cycling. *Transport reviews*, 36(1):9–27, 2016.



(a) Original DoS, $\overline{D}^{\mathrm{b}} = 4.41$, $P^{\mathrm{b}} = 16.49$ %

(b) Utilitarian DoS, $\overline{D}^{\mathrm{b}} = 4.36$, $P^{\mathrm{b}} = 18.05$ %.

(c) Sufficientarian DoS, $\overline{D}^{\mathrm{b}} = 4.31$, $P^{\mathrm{b}} = 19.48$ %.

(d) Original safety path insufficiency

(e) Utilitarian safety path insufficiency

(f) Sufficientarian safety path insufficiency

(g) Original bike travel time insufficiency

(h) Utilitarian bike travel time insufficiency

(i) Sufficientarian bike travel time insufficiency

(j) Original car travel time insufficiency

(k) Utilitarian car travel time insufficiency

(l) Sufficientarian car travel time insufficiency

Fig. 11: The top section reports the degree-of-separation (DoS) configurations together with their corresponding average cycling distance $\bar{D}^{\mathrm{b}}$ and bicycle mode share P^{b}, while the bottom panels illustrate the regional exposure-based path insufficiency, bicycle travel time insufficiency, and car travel time insufficiency, respectively. All optimization algorithms are applied with a exposure-based path insufficiency threshold of $\lambda = 40\%$ and a budget of $B \approx 0.15\, B_{\max}$.

[19] Peter Cox and Till Koglin. The politics of cycling infrastructure. *Policy, Bristol*, 2020.

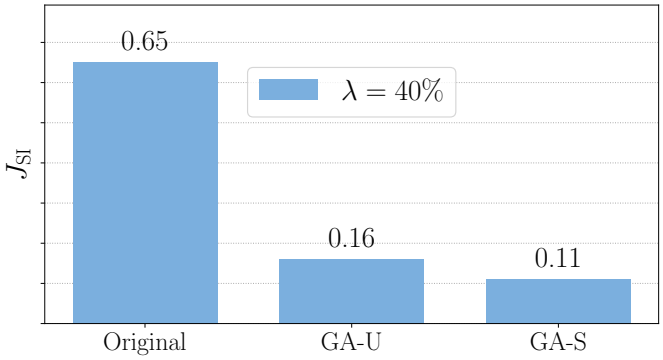


Fig. 12: Exposure-based path insufficiency comparison of upper level $\mathbb{U}_{\text{thr}}$ of Problem 3, and the color legend indicates the λ threshold.

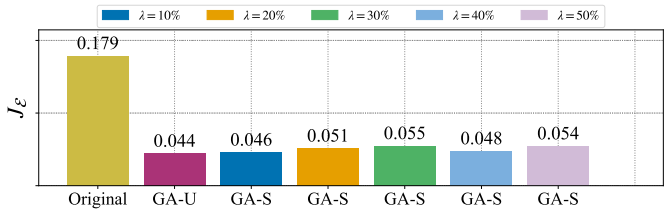


Fig. 13: On the top normalized cyclist exposure.

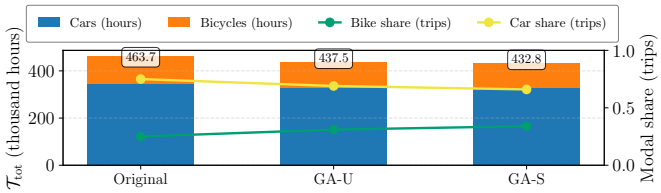


Fig. 14: Total system travel time ($\mathcal{T}_{\text{tot}}$).

[20] Carole Adam and Benoit Gaudou. An agent-based model of modal choice with perception biases and habits. *arXiv preprint arXiv:2406.02063*, 2024.

[21] Andres Monzon and Luis Alfredo Vega. How to measure the potential of transferring trips from car to public transport and soft modes—the case of madrid. In *Proceedings of the European Transport Conference, Strasbourg, France*, pages 18–20, 2006.

[22] Andres Monzon, Luis Alfredo Vega, and Maria Eugenia Lopez-Lambas. Potential to attract drivers out of their cars in dense urban areas. *European Transport Research Review*, 3(3):129–137, 2011.

[23] Mehrdad Bagheri, Milos N Mladenovic, Iisakki Kosonen, Jukka K Nurminen, Claudio Roncoli, and Antti Ylä-Jääski. A computational framework for revealing competitive travel times with low-carbon modes based on smartphone data collection. *Journal of Advanced Transportation*, 2020(1):4693750, 2020.

[24] Mehrdad Bagheri, Miloš N Mladenović, Iisakki Kosonen, and Jukka K Nurminen. Analysis of potential shift to low-carbon urban travel modes: a computational framework based on high-resolution smartphone data. *Sustainability*, 12(15):5901, 2020.

[25] John Stuart Mill. Utilitarianism. In *Seven masterpieces of philosophy*, pages 329–375. Routledge, 2016.

[26] Ken Binmore. Egalitarianism versus utilitarianism. *Utilitas*, 10(3):353–367, 1998.

[27] Rafael HM Pereira, Tim Schwanen, and David Banister. Distributive justice and equity in transportation. *Transport reviews*, 37(2):170–191, 2017.

[28] John Rawls. *A Theory of Justice*. Harvard University Press, Cambridge, MA, 1971.

[29] Karel Martens, Matan E Singer, and Aviv Lee Cohen-Zada. Equity in accessibility: Moving from disparity to insufficiency analyses. *Journal of the American Planning Association*, 88(4):479–494, 2022.

[30] Mauro Salazar, Sara Betancur Giraldo, Fabio Paparella, and Leonardo Pedroso. On accessibility fairness in intermodal autonomous mobility-on-demand systems. *IFAC-PapersOnLine*, 58(10):327–333, 2024.

[31] Julianno Amorim and João de Abreu e Silva. Comparing the application of different justice theories in equity analysis of transit projects: A case study of the lisbon metro circular line. *Journal of Transport and Land Use*, 17(1):21–40, 2024.

[32] Mauro Salazar, Sara Betancur Giraldo, Fabio Paparella, Leonardo Pedroso, and Karel Martens. Mobilizing transport justice: a sufficientarian optimization framework for intermodal mobility systems. *npj Sustainable Mobility and Transport*, 2(1):43, 2025.

[33] Manuel Campero Jurado, Carlos Canudas-De-Wit, Giovanni de Nunzio, and Mauro Salazar. On transport justice and safety in bicycle network design optimization. In *CDC 2025-IEEE 64th Conference on Decision and Control*, pages 1–6. IEEE, 2025.

[34] Esta Qiu, David Rey, and Travis Waller. Multi-period bus frequency optimization and fleet rebalancing based on equity over time. *Transportation*, pages 1–51, 2025.

[35] Manuel Campero Jurado, Giovanni de Nunzio, and Carlos Canudas-De-Wit. Safety-oriented design of cycling infrastructure minimizing impact on vehicular traffic. In *28th International Conference on Intelligent Transportation Systems*, 2025.

[36] GTLVille Team. Gtlville, 2024. Accessed: 2024-26-10.

[37] Manuel Campero Jurado, Carlos Canudas-de Wit, and Giovanni De Nunzio. A graph-based approach to the topological optimization of cycling networks for the improvement of safety and comfort of cyclists. *IEEE Transactions on Intelligent Transportation Systems*, 2025.

[38] Crispin HV Cooper. Using spatial network analysis to model pedal cycle flows, risk and mode choice. *Journal of transport geography*, 58:157–165, 2017.

[39] Manuel Campero Jurado, Mauro Salazar, Giovanni De Nunzio, and Carlos Canudas de Wit. Improving urban cycling safety and comfort through optimized infrastructure upgrades. *In 23rd European Control Conference (ECC)*, 2025.

[40] International Road Assessment Programme. Cyclerap, 2024. Accessed: 2024-07-10.

[41] Johannes Mesimäki and Juha Luoma. Near accidents and collisions between pedestrians and cyclists. *European transport research review*, 13(1):38, 2021.

[42] Sven Maerivoet and Bart De Moor. Transportation planning and traffic flow models. *arXiv preprint physics/0507127*, 2005.

[43] United States. Bureau of Public Roads. *Traffic assignment manual for application with a large, high speed computer*, volume 37. US Department of Commerce, Bureau of Public Roads, Office of Planning, Urban ..., 1964.

[44] M. J. Beckmann. A lagrangian multiplier rule in linear activity analysis and some of its applications. Technical Report Paper No. 2054, Cowles Commission for Research in Economics, 1952.

[45] Martin Beckmann, Charles B. McGuire, and Christopher B. Winsten. Studies in the economics of transportation. Research report, Cowles Commission for Research in Economics, 1956.

[46] Pivithuru Kalpana Nanayakkara, Nano Langenheim, Irene Moser, and Marcus White. Do safe bike lanes really slow down cars? a simulation-based approach to investigate the effect of retrofitting safe cycling lanes on vehicular traffic. *International journal of environmental research and public health*, 19(7):3818, 2022.

[47] Geoff Boeing. Osmnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks. *Computers, Environment and Urban Systems*, 65:126–139, 2017.

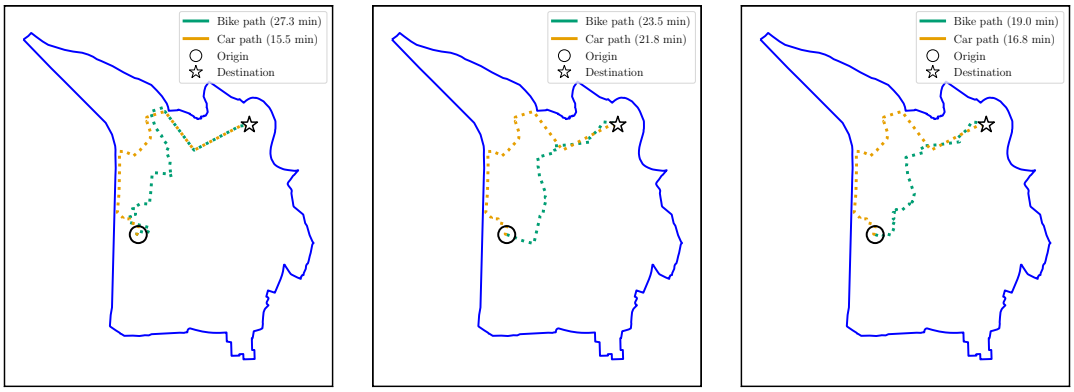
[48] Pascal Oger and Sandra Bouvet. Aller travailler à vélo, une pratique très urbaine. *Insee flash Auvergne-Rhône-Alpes*, 2017. ©Insee 2017.

[49] Agence d'urbanisme de la région grenobloise. Enquête mobilité 2019-2020: Chiffres clés. Technical report, SMMAG and CEREMA, 2020.

[50] Cycling Industries Europe and Megabits. The potential for bike sharing in europe: Benchmarking 148 cities. Project report, Shared Ambition, Interreg North Sea Region, 2024. Available at: https://cyclingindustries.com/.

[51] Clara Glachant and Frauke Behrendt. Negotiating the bicycle path: A study of moped user stereotypes and behaviours in the netherlands. *Transportation Research Part F: Traffic Psychology and Behaviour*, 107:301–320, 2024.

[52] Avi Parsha and Karel Martens. Social identity and cycling among women: The case of tel-aviv-jaffa. *Transportation research part F: traffic psychology and behaviour*, 89:1–15, 2022.



(a) Original DoS **(b)** utilitarian DoS **(c)** Sufficientarian DoS

Fig. 15: Grenoble configuration paths under different optimizations algorithms.

[53] Yasser Amiour, E Owen D Waygood, and Pauline EW van den Berg. Objective and perceived traffic safety for children: a systematic literature review of traffic and built environment characteristics related to safe travel. *International journal of environmental research and public health*, 19(5):2641, 2022.

[54] Stefan Fuest, Mariana Batista, Frauke Luise Berghoefer, Morten Flesser, Bhagya Shrithi Grandhi, Felix Spühler, Monika Sester, and Mark Vollrath. I bet you feel safe! assessing cyclists' subjective safety by objective scores. *Journal of Urban Mobility*, 4:100066, 2023.

[55] Sergio A Useche, Luis Montoro, Jaime Sanmartin, and Francisco Alonso. Healthy but risky: A descriptive study on cyclists' encouraging and discouraging factors for using bicycles, habits and safety outcomes. *Transportation research part F: traffic psychology and behaviour*, 62:587–598, 2019.

[56] CEREMA, Bron, France. *Aménagements provisoires pour les piétons : tester pour aménager durablement*, 2020. Cité des mobilités - 25 avenue François Mitterrand - CS 92803 - F-69674 Bron Cedex, Tél. +33 (0)4 72 14 30 30.

[57] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Kopf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. Pytorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems 32*, pages 8024–8035. Curran Associates, Inc., 2019.

[58] Alice Brauns, Alain Dalouche, Axel Lambert, Catherine Pilon, and Laura Youkana. Le marché aménagements cyclables. Printed document, 2024.

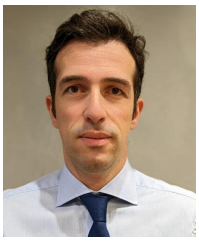
[59] Grenoble-Alpes-Métropole. Budget 2025: L'investissement renforcé malgré l'étau budgétaire du gouvernement, 2025. Contact presse: Randa Berbouche, randa.berbouche@grenoblealpesmetropole.fr, 06 82 85 02 08.



Manuel Campero Jurado Manuel Campero Jurado received the B.S. degree in applied mathematics in 2019 and the M.S. degree in mathematics in 2021, both from the Autonomous University of the State of Hidalgo, Mexico, and the Ph.D. degree from the Grenoble Institute of Technology, Grenoble, France in 2026. He is currently a postdoctoral researcher at IFP Energies nouvelles (IFPEN), Rueil-Malmaison, France. His research interests include modeling and optimization of transport networks, with a focus on low-carbon mobility.



Carlos Canudas de Wit received the B.Sc. degree in electronics and communications from the Monterrey Institute of Technology, Mexico, in 1980, and the M.Sc. and Ph.D. degrees in automatic control from the Grenoble Institute of Technology, France, in 1984 and 1987, respectively. He is a Directeur de recherche (Senior Researcher) at CNRS, Grenoble, where he leads the NeCS team (GIPSA-Lab and INRIA) on networked controlled systems. He is an IFAC Fellow, has served as Associate Editor for several IEEE journals, and was President of the European Control Association from 2013 to 2015. He held the ERC Advanced Grant Scale-FreeBack from 2016 to 2021. His research focuses on electric vehicles, optimal control, traffic modeling, and large-scale networks.



Giovanni De Nunzio received his B.Sc. and M.Sc. degrees in Information and Automation Engineering from the University of L'Aquila, Italy, in 2007 and 2010, respectively, and the Ph.D. degree in Automatic Control from the Grenoble Institute of Technology (Grenoble INP), France, in 2015. He is currently a Research Engineer and a Research and Innovation Project Leader with IFP Energies nouvelles, Lyon, France. His research activities focus on modeling and control of transportation and mobility systems, advanced driver-assistance systems, graph theory, optimization, data science, and artificial intelligence.



Mauro Salazar received the B.Sc. and M.Sc. degrees in Mechanical Engineering from ETH Zürich, Switzerland, in 2012 and 2015, respectively, and carried out his Master's thesis at the Automatic Control Laboratory of EPFL. He received the Ph.D. degree in Mechanical Engineering from ETH Zürich in 2019. He is an Associate Professor with the Control Systems Technology Section, Eindhoven University of Technology (TU/e), and is co-affiliated with the Eindhoven AI Systems Institute (EAISI). Before joining TU/e, he was a Postdoctoral Scholar with the Autonomous Systems Laboratory at Stanford University. His research focuses on optimization models and methods for the design and control of cyber-socio-technical systems, with a strong focus on sustainable mobility. Both his Master's and Ph.D. theses were recognized with the ETH Medal, and his papers received the Best Student Paper Award at the 2018 IEEE Intelligent Transportation Systems Conference and the 2022 European Control Conference